



Management of Energy Efficiency and Safe Operation for Multi-Actuated Ground Vehicles on DWPT Roads

MVC 2027 Organizers

Binh-Minh Nguyen¹, Facundo Aguilera²

¹The University of Tokyo, ²National University of Río Cuarto

IEEE VTS Motor Vehicles

Ricardo de Castro³

³University of California, Merced

Agenda

☒ Introduction

☐ Motivation

☐ Problem setting of MVC2027

☐ MATLAB/Simulink files

☐ Demonstration of MESO strategies

☐ GitHub repository

☐ Registration and Submission

Introduction to MVC (Ricardo de Castro)

Agenda

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☒ **Motivation**

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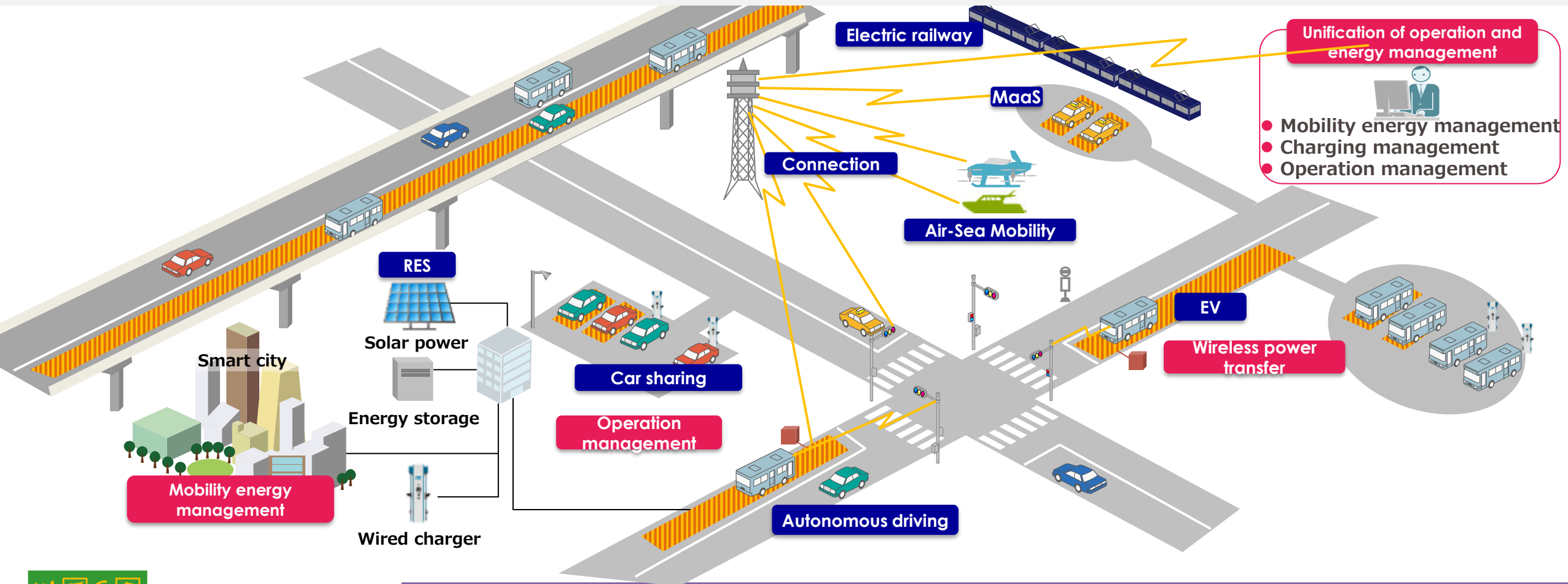
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The future of e-Mobility

The transportation sector is undergoing a fundamental transformation driven by the urgent need to reduce carbon emissions and develop sustainable urban mobility systems.

- ❖ Emerging technologies such as *autonomous driving and motion control*, *optimal energy management and operation management*, and *dynamic wireless power transfer* are shaping the future of mobility.
- ❖ The development and integration of these systems demand not only technological innovation but also a *transdisciplinary approach* integrating expertise in electrical, automotive, control, mechanical, material engineering, etc.



Our work toward the future of e-Mobility

Offline Energy Estimation (START)

- Psth speed trajectories clustering.
- Stochastic traffic flow model using GMM.
- Add up the energy consumption of each signal pair at each average speed within the prediction horizon

Online STO

- (Traffic signal detection) $i = 0$
- Cluster selection
- Stochastic learning model for GMM is determined
- Energy estimation for terminal cost for optimization problem
- Speed trajectory optimization by DP using probability distribution of v_i
- Final traffic signal? Yes → END, No → $i = i + 1$

Y. Hosomi et al, "Data-driven Energy-optimized Speed Trajectory for Urban Driving Electric Vehicles Utilizing Traffic Flow Estimation," IEEE Transactions on Transportation Electrification (2025).

Speed trajectory generation

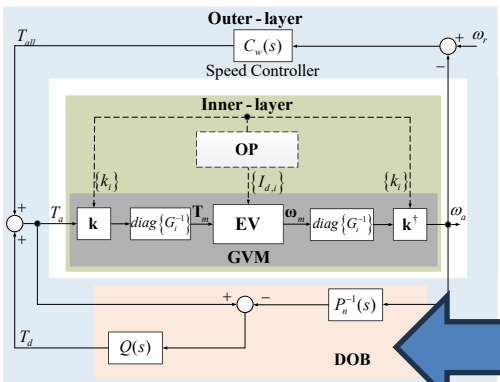


Slippery surface

In-wheel motor

B. -M. Nguyen et al, "Slip Control for IWM Vehicles Based on Hierarchical LQR," Control Engineering Practice, Vol. 93, 104179 (2019).

Torque allocation



Outer-layer

Speed Controller

Inner-layer

OP

GMM

DOB

B. -M. Nguyen et al, "Double-Layer Energy Management for Multi-Motor Electric Vehicles," IEEE Transactions on Vehicular Technology, Vol. 72, No. 7, pp. 8623-8635 (2023).

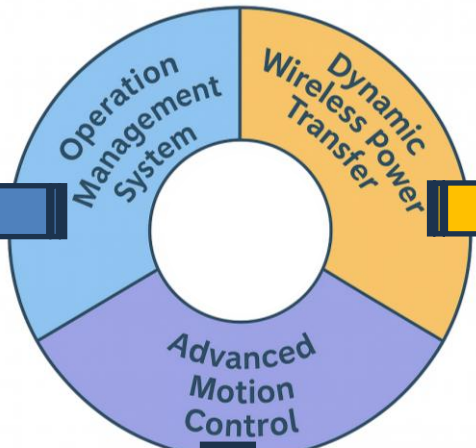
Friction force control



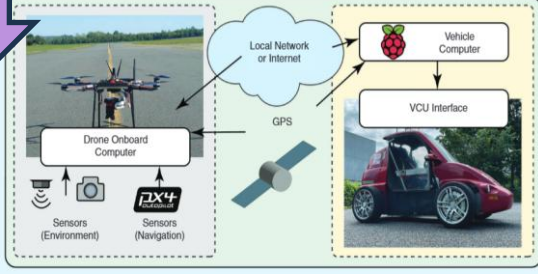
Motor drive via pulley & belt

D. Lazzarini et al, "On the Simulation and Experimental Analysis of Imitation Learning for Preview-Based Torque Vectoring Control," Vehicle System Dynamics (2026).

Advanced Motion Control



Road condition estimation



Local Network or Internet

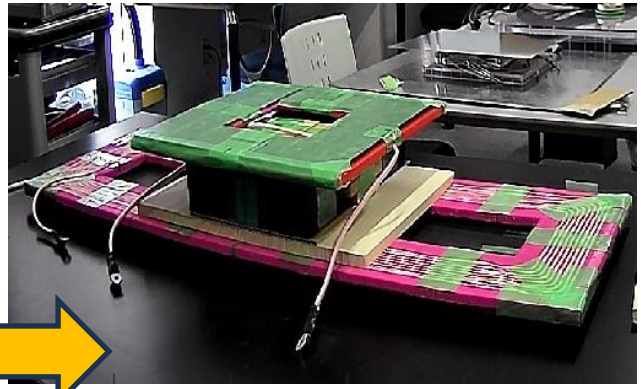
Vehicle Computer

VCU Interface

GPS

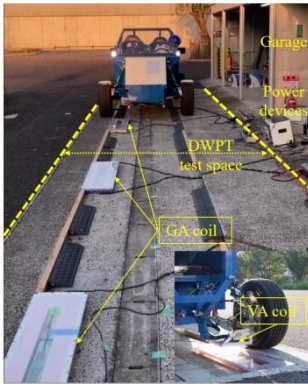
V. Beliautsou et al, "Drone-to-Vehicle Integration of Data: Design Concept and Application to Vehicle Automated System," IEEE Vehicular Technology Magazine (2025).

Coil design



T. Koishi et al, "Receiving Side Current-Based Lateral Misalignment Estimation and Automated Steering Control for Dynamic Wireless Power Transfer," IEEE Journal of Emerging and Selected Topics in Industrial Electronics, Vol. 6, No. 1, pp. 41-52 (2024).

Position control



Garage

Power devices

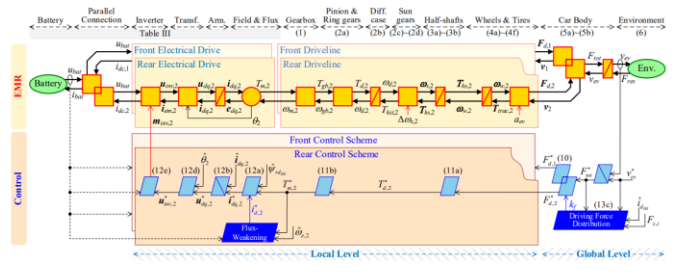
DWPT test space

GA coil

VA coil

A. -T. Nguyen et al, "An Integrated Modeling Framework for Motion Control and Energy Management in Multi-Motor Electric Vehicles," IEEE Open Journal of Vehicular Technology (2025).

EMR Modeling of Vehicle



EMR

Control

Local Level

Global Level

Example: Integration of IWM and EPS Control

Without Control



→ Spin on low friction surface

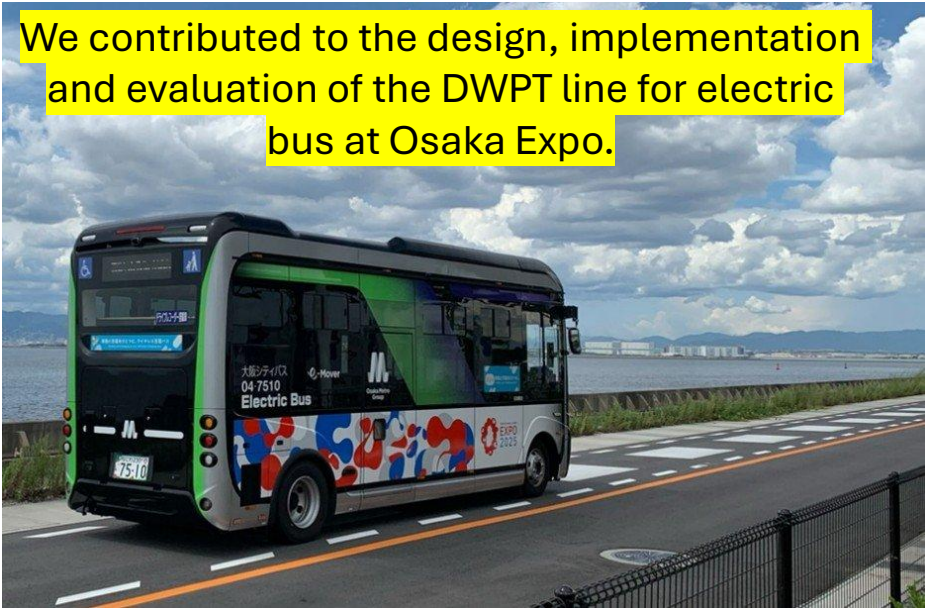
With Control



→ Stable motion is maintained

Our DWPT Development

We contributed to the design, implementation and evaluation of the DWPT line for electric bus at Osaka Expo.



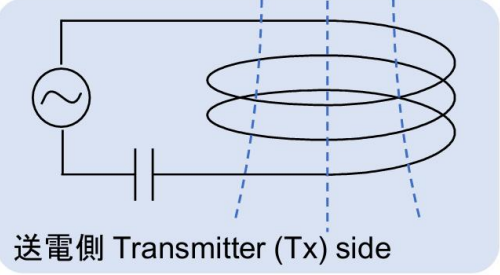
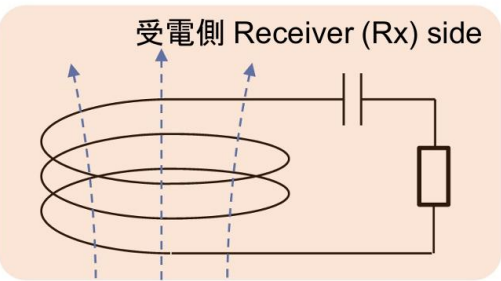
www.k.u-tokyo.ac.jp/information/category/press/11765.html



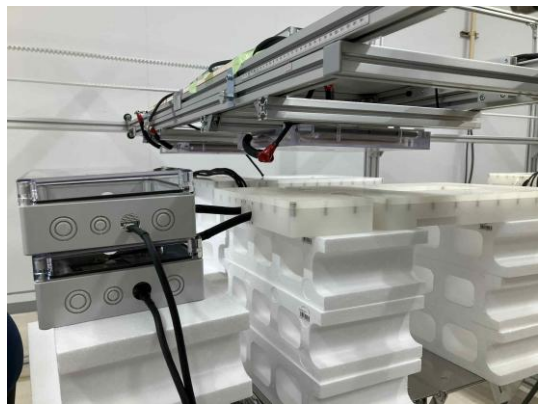
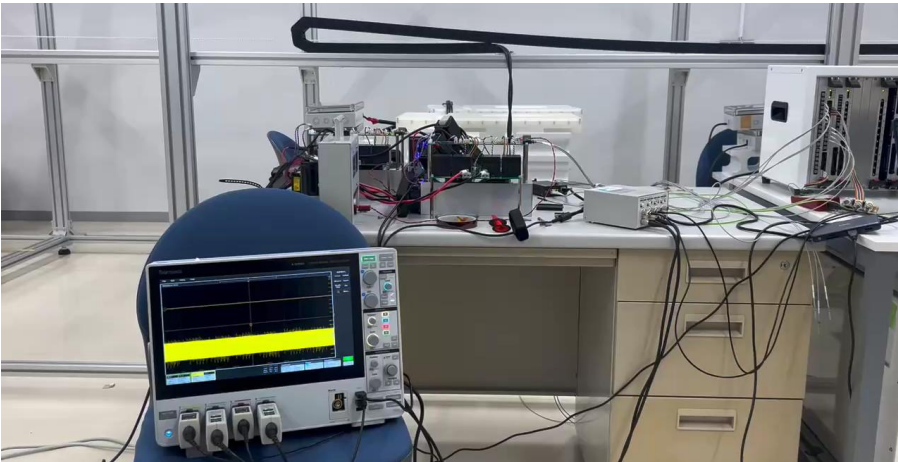
DWPT section at Kashiwanoha, Kashiwa City, Chiba Prefecture, Japan



Faraday: "Time varying magnetic flux induces electromotive force"



Ampere: "Electric current creates magnetic field around it"



Goal of IEEE VTS MVC 2027

Management of Energy Efficiency and Safe Operation for Multi-Actuated Ground Vehicles on DWPT Roads

- ❖ Enhance the understanding of future e-Mobility concepts, enabling technologies, and key challenges.
- ❖ Demonstrate the need for a *transdisciplinary approach* to developing next-generation e-Mobility systems.
- ❖ Provide participants and the broader community with a multipurpose simulator for vehicle dynamics education, evaluation of research ideas, and collaborative research, etc.
- ❖ Attract participants from diverse academic and professional communities.

Remark: The simulation is based on a real experimental vehicle operated at the e-Mobility and Control Laboratory, The University of Tokyo. To ensure a feasible problem setting and maintain reasonable computational cost, certain dynamics and components of the real system are simplified or neglected. For example, the dynamics of the energy storage system and the power-electronics control, including coil detection and current control for the DWPT system, are not explicitly modeled. Nevertheless, the resulting model remains sufficiently detailed to provide a challenging simulation environment. The simulator has been developed using MATLAB R2022b, with a sampling time of 0.0002 s.



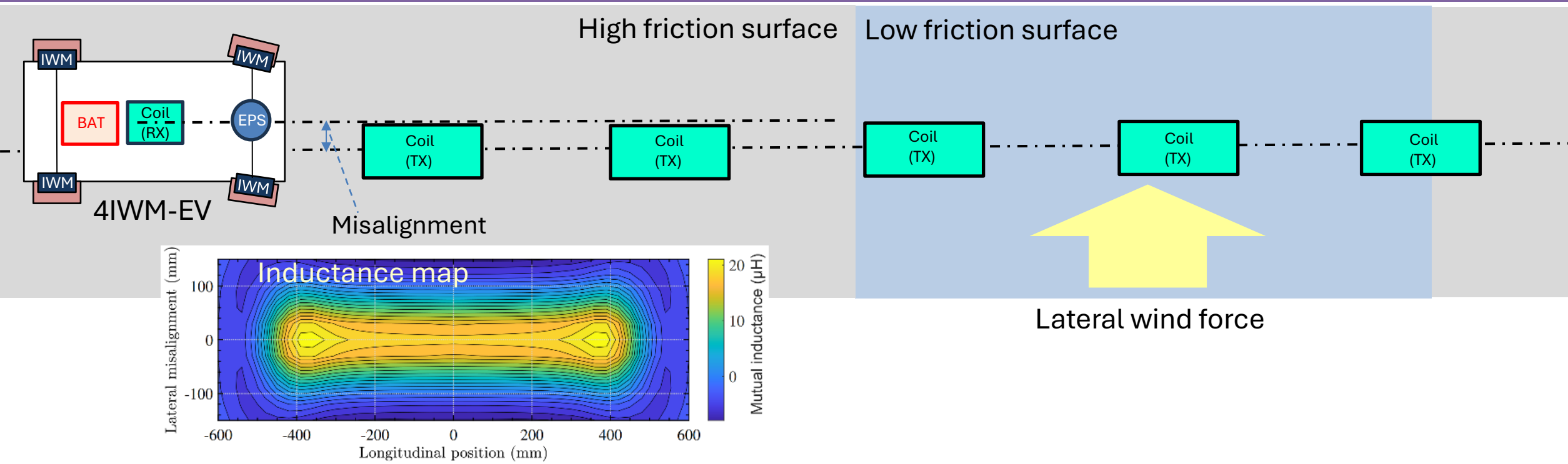
Timeline

- ❖ 2025/10~2026/03: Problem formulation and simulator development
- ❖ 2026/04~2026/09: Preparation and final submission of MVC 2027 paper to VPPC 2026
- ❖ 2026/10: Presentation at VPPC 2026 and collection of feedback from research community
- ❖ 2026/10: Public release of MVC2027 on the website
- ❖ 2026/11~2027/01: Participant registration
- ❖ 2027/02: Submission of proposals by participants
- ❖ 2027/03: Evaluation and scoring of the submitted proposals

Agenda

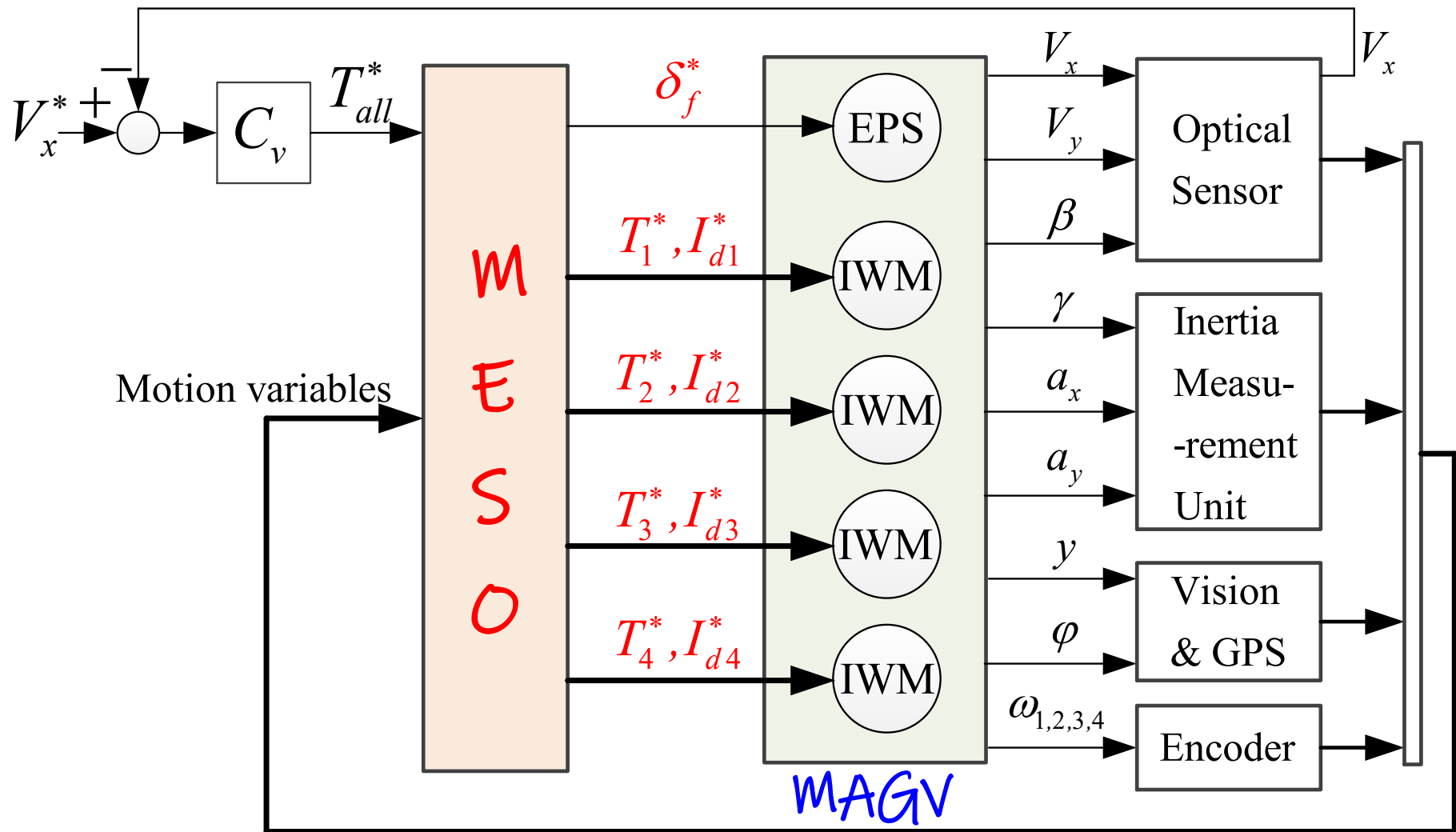
- ❑ Introduction
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Electric Vehicle and Driving Scenario



- The electric vehicle is actuated by **four** in-wheel-motor and **one** front steering motor. It runs on a *straight* road. The speed pattern and speed controller are given.
- The vehicle may encounter a *low-friction* road surface and strong *lateral wind disturbances*.
- The DWPT is only available if the longitudinal and lateral misalignments are between certain thresholds. For instance, if the lateral misalignment exceeds 70mm, DWPT is shut down.

Block Diagram of MESO



Objectives

The participants are invited to design a strategy that generates {torque distribution ratios, motor d-axis currents, front steering angle} to attain multiple objectives:

(1) Drivetrain efficiency: Minimize the energy consumption:

$$\Phi_1 = \int_0^{t_{end}} P_{all} dt = \int_0^{t_{end}} \sum_{i=1}^4 (P_{m,i} + P_{cu,i} + P_{fe,i}) dt$$

(2) Safe traction: Minimize the wheel slip ratios:

$$\Phi_2 = \int_0^{t_{end}} \sum_{i=1}^4 \lambda_i^2 dt$$

(3) DWPT efficiency: Minimize DWPT efficiency penalty:

$$\Phi_3 = \int_0^{t_{end}} (1 - \eta_{DWPT})^2 dt$$

(4) Driving comfort: Minimize lateral acceleration:

$$\Phi_4 = \int_0^{t_{end}} a_y^2 dt$$

Scoring function: $\Phi = \sigma_1 \Phi_1 + \sigma_2 \Phi_2 + \sigma_3 \Phi_3 + \sigma_4 \Phi_4$

The weighting coefficients account for both the characteristic magnitudes and the relative importance of the four evaluation criteria.

Constraints

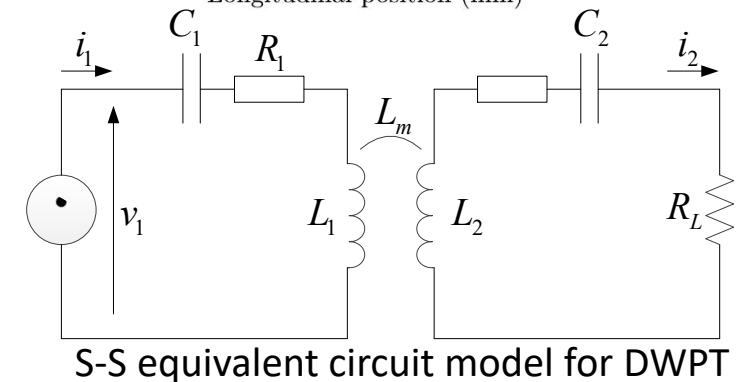
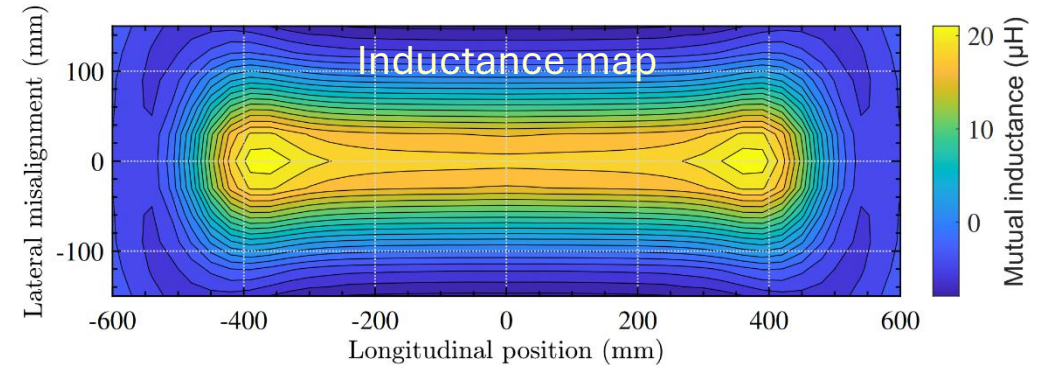
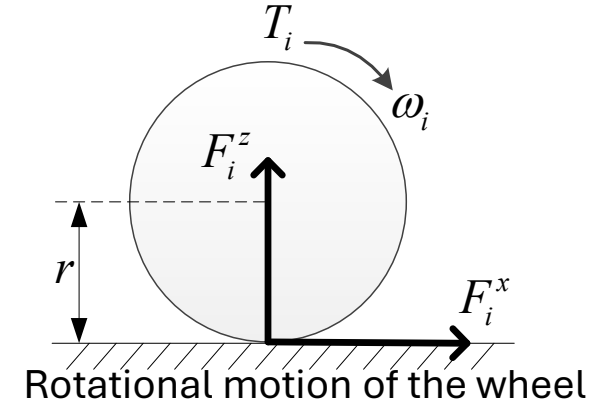
(1) **Total torque command:** $\sum_{i=1}^4 T_i^* = T_{all}^*$, where T_{all}^* is given by the velocity controller.

(2) **Motor torque limits:** $|T_i^*| \leq T_{max,i}, \forall i \in [1,4]$.

(3) **Friction limit circle:** $\sqrt{(F_i^x)^2 + (F_i^y)^2} \leq \mu F_i^z$,

(4) **Coil misalignment:** Wireless power transfer is enabled only when $|\Delta_x| \leq 380$ mm and $|\Delta_y| \leq 70$ mm, where Δ_x and Δ_y denote the longitudinal and lateral misalignments between TX and RX coils, respectively. This is to ensure sufficient mutual inductance and efficiency during power transfer operation:

$$\eta_{DWPT} = \frac{(\omega_0 L_m)^2}{R_1 R_2 + R_1 R_L + (\omega_0 L_m)^2} \frac{R_L}{R_2 + R_L}$$



Agenda

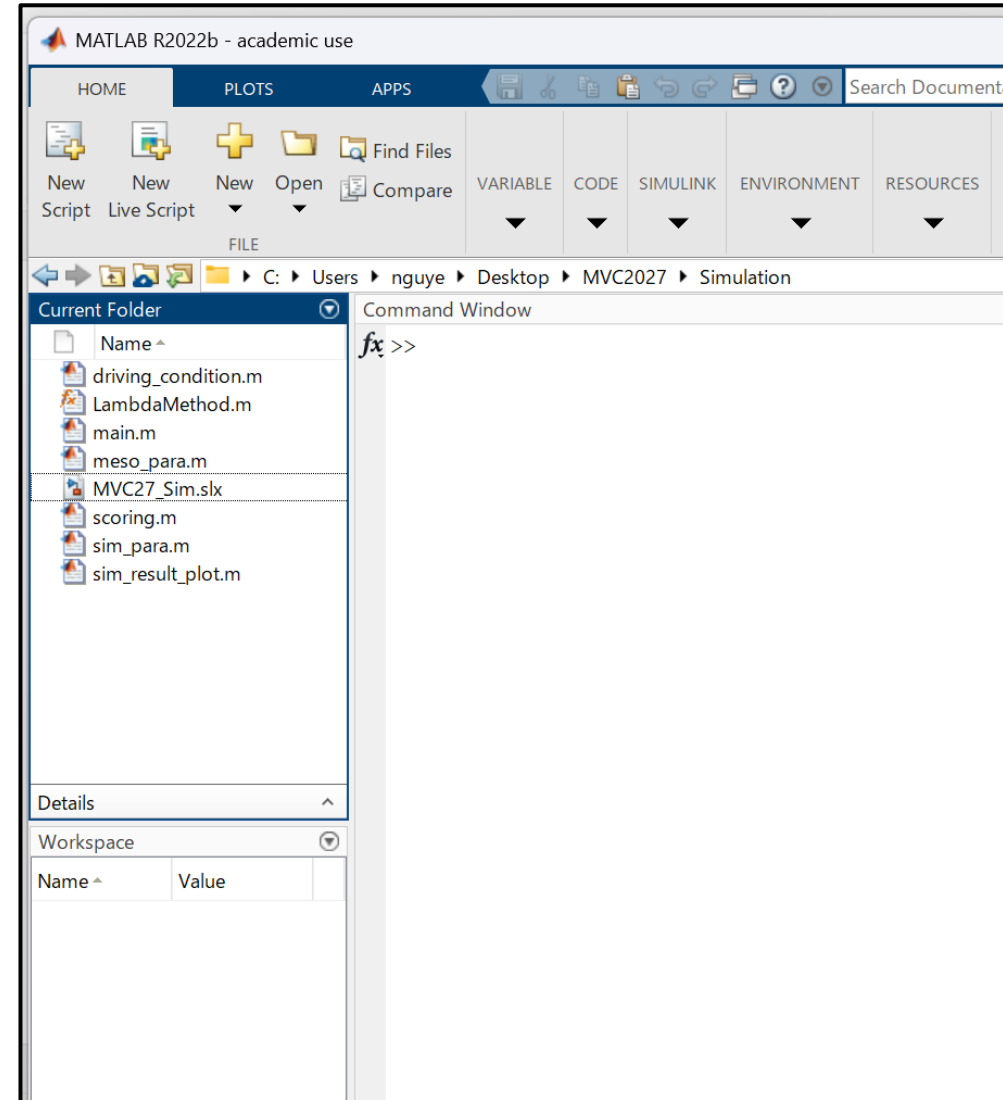
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Simulation details

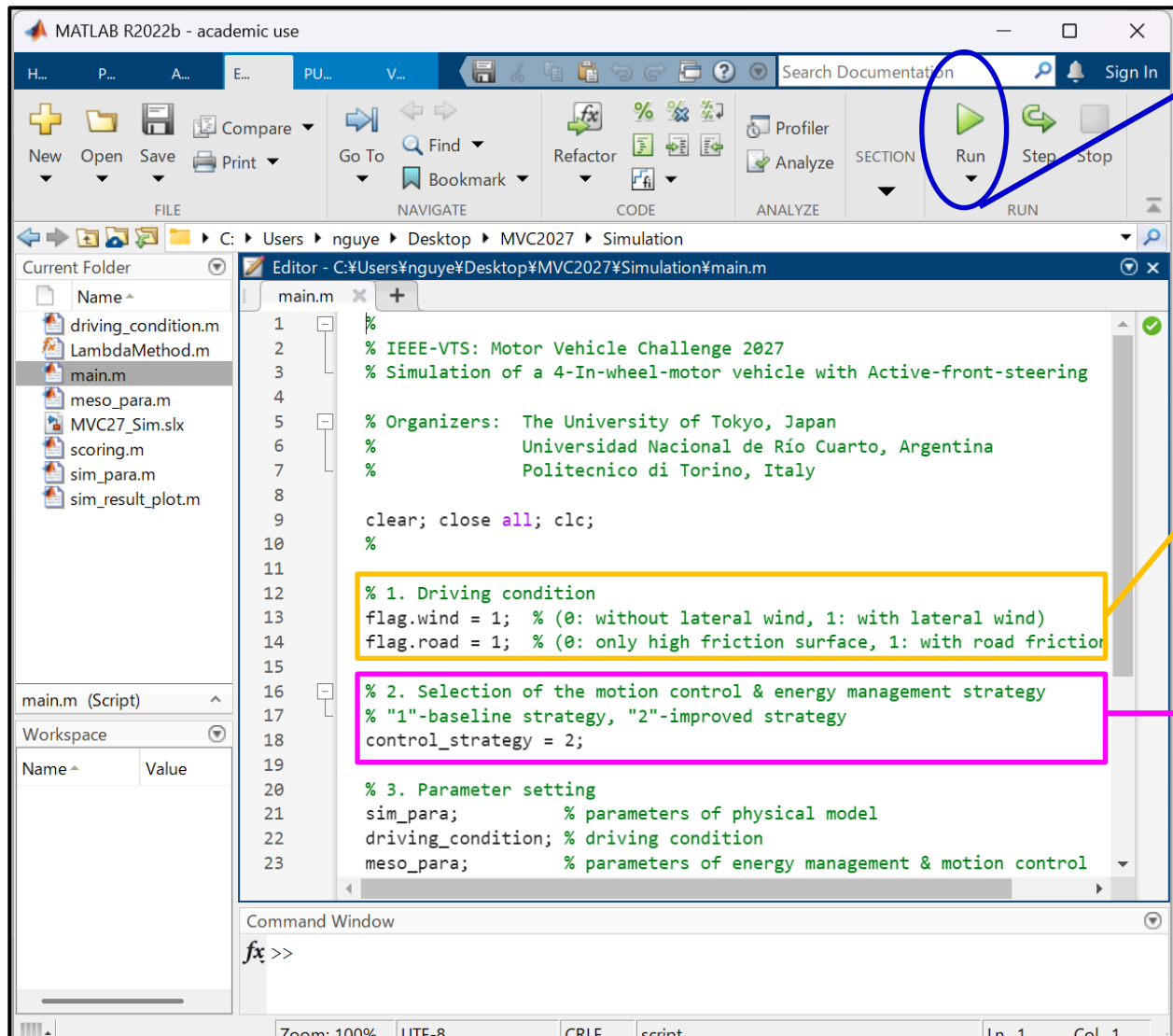
❑ The Simulator has been developed in the MATLAB/Simulink environment (**R2022b**).

❑ Files:

- **main.m** (to run overall simulation)
- **sim_para.m** (parameters of the electric vehicle)
- **meso_para.m** (parameters of the strategies) → *to be designed*
- **driving_condition.m** (setting of road friction & wind force)
- **LambdaMethod.m** (to calculate tire forces using Magic Formula)
- **sim_result_plot.m** (to plot the simulation results)
- **scoring.m** (to calculate the score)
- **MVC27_Sim.slx** (simulation file) → *to be designed*



main.m



❑ “main.m” is used to run the overall simulation.

❑ Before running “main.m”, it is essential to select the driving condition and the motion control & energy management strategy.

❑ For instance, the vehicle runs on the road with lateral wind disturbance and road friction change if we select:
$$\text{flag.wind} = 1; \text{flag.road} = 1;$$

❑ The baseline strategy is selected when $\text{control_strategy} = 1;$ whereas the improved strategy is selected when $\text{control_strategy} = 2;$

sim_para.m

```
1 %
2 % Parameter of the simulator
3
4 % 1. Constant
5 g      = 9.81;      % acceleration of gravity [m/s^2]
6 pi     = 3.14;      % number pi
7 eps    = 0.05;      % small value for anti-division-by-zero i
8 %
9
10 %
11 % 2. Vehicle body parameter
12 Vehicle.M = 880;      % total mass [kg]
13 Vehicle.I = 617.0;    % yaw moment of inertia [kg.m^2]
14 Vehicle.lf = 0.988;   % distance from CG to front axle [m]
15 Vehicle.lr = 0.712;   % distance from CG to rear axle [m]
16 Vehicle.l = Vehicle.lf + Vehicle.lr; % wheelbase [m]
17 Vehicle.d = 1.3;      % track width
18 Vehicle.df = Vehicle.d; % track width (front) [m]
19 Vehicle.dr = Vehicle.d; % track width (rear) [m]
20 Vehicle.hg = 0.46;    % height of CG [ m ]
21 Vehicle.mu0 = 0.0128; % rolling-friction resistance
22 Vehicle.rho = 1.205;  % air density
23 Vehicle.Cd = 0.48;    % drag coefficient
24 Vehicle.Afpa = 2.4;   % frontal projected area [m^2]
25 Vehicle.Nf = (Vehicle.M/2)*g*Vehicle.lr/Vehicle.l; % nominal load (front) [N]
26 Vehicle.Nr = (Vehicle.M/2)*g*Vehicle.lf/Vehicle.l; % nominal load (rear) [N]
```

❑ “sim_para.m” defines the physical parameters of the electric vehicle model.

❑ The parameters are categorized as follows:

% 1. Constant

% 2. Vehicle body parameter

% 3. Wheel parameter

% 4. Steering model

% 5. Road condition

% 6. Motor parameter

% 7. Control parameter

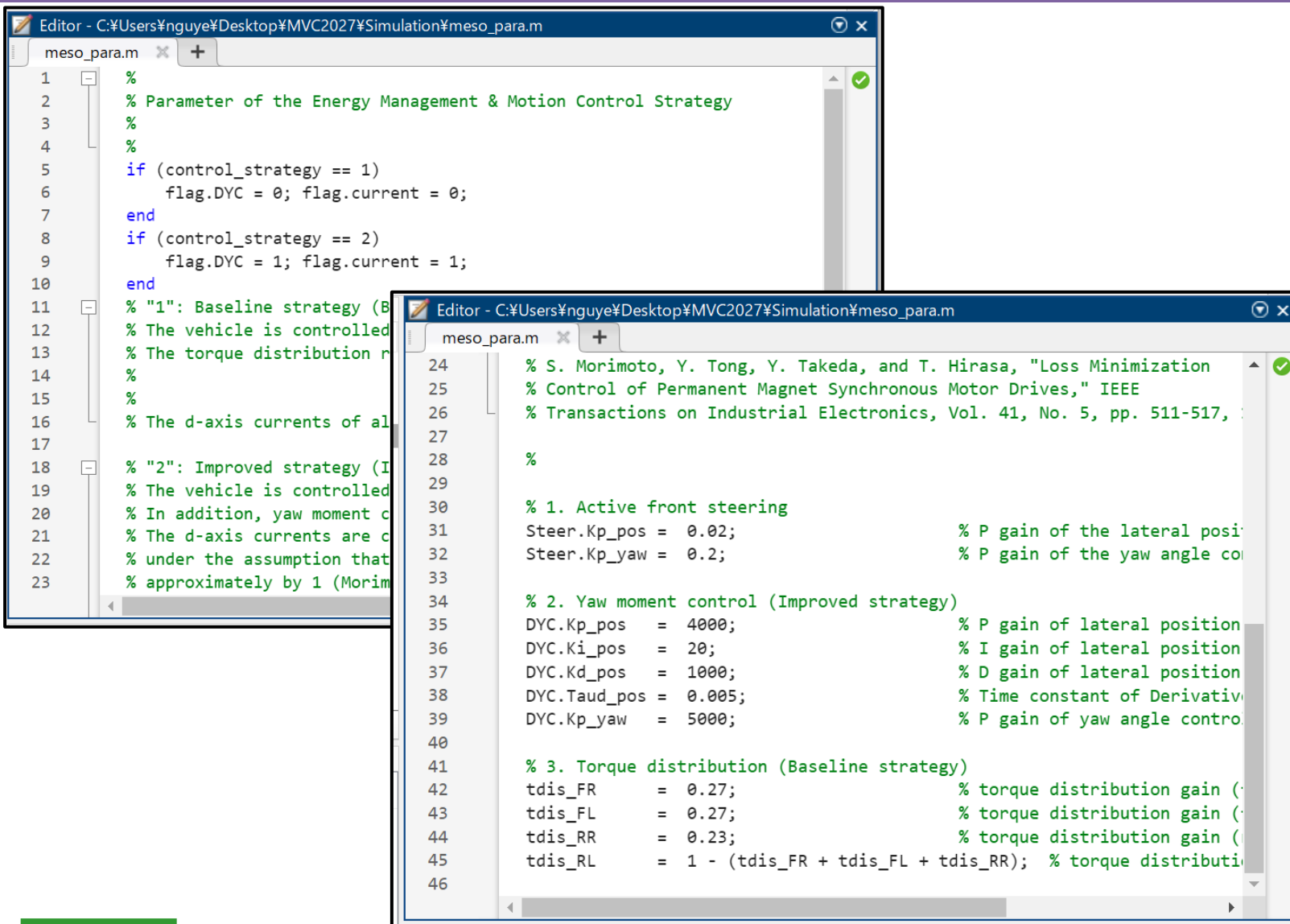
% 8. Simulation setting

% 8.1. Simulation time and initial

% 8.2. Lateral wind disturbance

% 9. Dynamic wireless power transfer

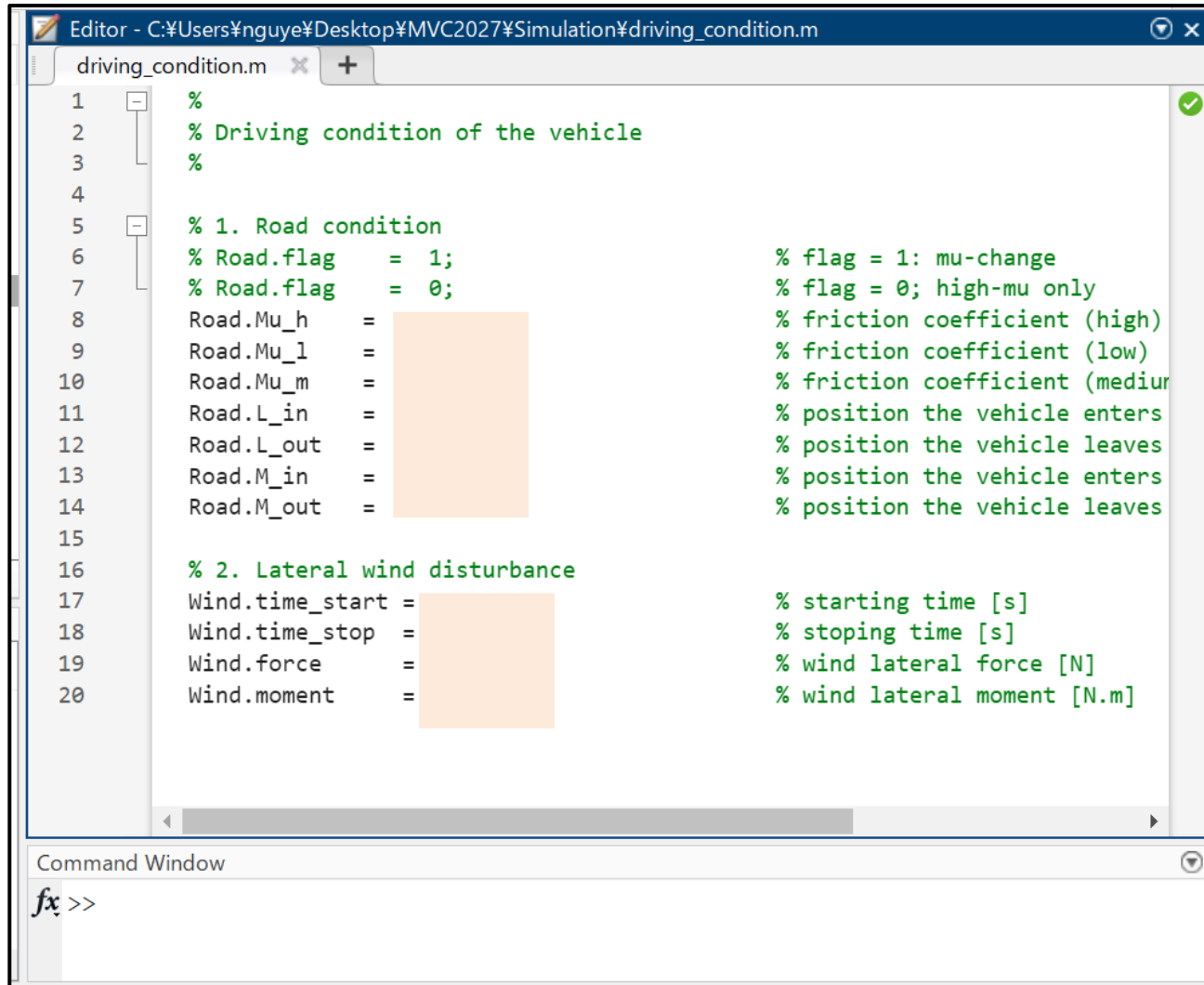
meso_para.m



```
1 %  
2 % Parameter of the Energy Management & Motion Control Strategy  
3 %  
4 %  
5 if (control_strategy == 1)  
6     flag.DYC = 0; flag.current = 0;  
7 end  
8 if (control_strategy == 2)  
9     flag.DYC = 1; flag.current = 1;  
10 end  
11 % "1": Baseline strategy (Baseline)  
12 % The vehicle is controlled by the torque distribution law  
13 % The torque distribution law is used to distribute the torque  
14 %  
15 %  
16 % The d-axis currents of all motors are controlled by the  
17 %  
18 % "2": Improved strategy (Improved)  
19 % The vehicle is controlled by the torque distribution law  
20 % In addition, yaw moment control is used to control the yaw  
21 % The d-axis currents are controlled by the torque distribution  
22 % under the assumption that the torque distribution law is  
23 % approximately by 1 (Morimoto, Y. Tong, Y. Takeda, and T. Hirasaka, "Loss Minimization  
24 % S. Morimoto, Y. Tong, Y. Takeda, and T. Hirasaka, "Loss Minimization  
25 % Control of Permanent Magnet Synchronous Motor Drives," IEEE  
26 % Transactions on Industrial Electronics, Vol. 41, No. 5, pp. 511-517, 1994)  
27 %  
28 %  
29 %  
30 % 1. Active front steering  
31 Steer.Kp_pos = 0.02; % P gain of the lateral position  
32 Steer.Kp_yaw = 0.2; % P gain of the yaw angle control  
33  
34 % 2. Yaw moment control (Improved strategy)  
35 DYC.Kp_pos = 4000; % P gain of lateral position  
36 DYC.Ki_pos = 20; % I gain of lateral position  
37 DYC.Kd_pos = 1000; % D gain of lateral position  
38 DYC.Taud_pos = 0.005; % Time constant of Derivative  
39 DYC.Kp_yaw = 5000; % P gain of yaw angle control  
40  
41 % 3. Torque distribution (Baseline strategy)  
42 tdis_FR = 0.27; % torque distribution gain (Front Right)  
43 tdis_FL = 0.27; % torque distribution gain (Front Left)  
44 tdis_RR = 0.23; % torque distribution gain (Rear Right)  
45 tdis_RL = 1 - (tdis_FR + tdis_FL + tdis_RR); % torque distribution  
46
```

- ❑ “meso_para.m” is used to define the parameters of the energy management and motion control strategy.
- ❑ For instance, it consists of the parameters of the following algorithms:
 - ✓ Active front steering control
 - ✓ Yaw moment control
 - ✓ Torque distribution law

driving_condition.m



```
Editor - C:\Users\nguye\Desktop\MVC2027\Simulation\driving_condition.m
driving_condition.m
1 %
2 % Driving condition of the vehicle
3 %
4
5 % 1. Road condition
6 % Road.flag = 1; % flag = 1: mu-change
7 % Road.flag = 0; % flag = 0; high-mu only
8 Road.Mu_h = % friction coefficient (high)
9 Road.Mu_l = % friction coefficient (low)
10 Road.Mu_m = % friction coefficient (medium)
11 Road.L_in = % position the vehicle enters
12 Road.L_out = % position the vehicle leaves
13 Road.M_in = % position the vehicle enters
14 Road.M_out = % position the vehicle leaves
15
16 % 2. Lateral wind disturbance
17 Wind.time_start = % starting time [s]
18 Wind.time_stop = % stoping time [s]
19 Wind.force = % wind lateral force [N]
20 Wind.moment = % wind lateral moment [N.m]

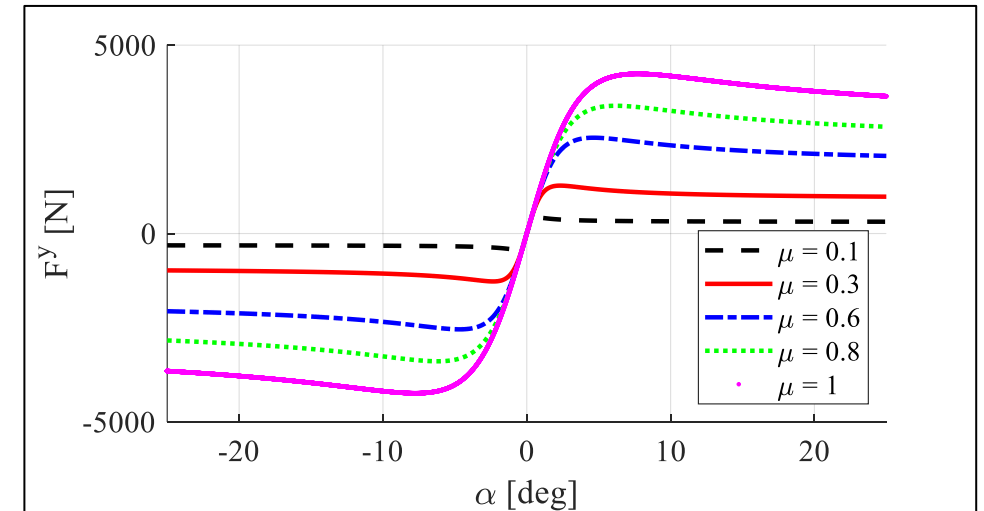
Command Window
fx >>
```

- ❑ “driving_condition.m” is used to define the road conditions and specify the lateral wind force disturbance.
- ❑ The control strategy must not directly access information and parameters defined by “driving_condition.m.”

LambdaMethod.m

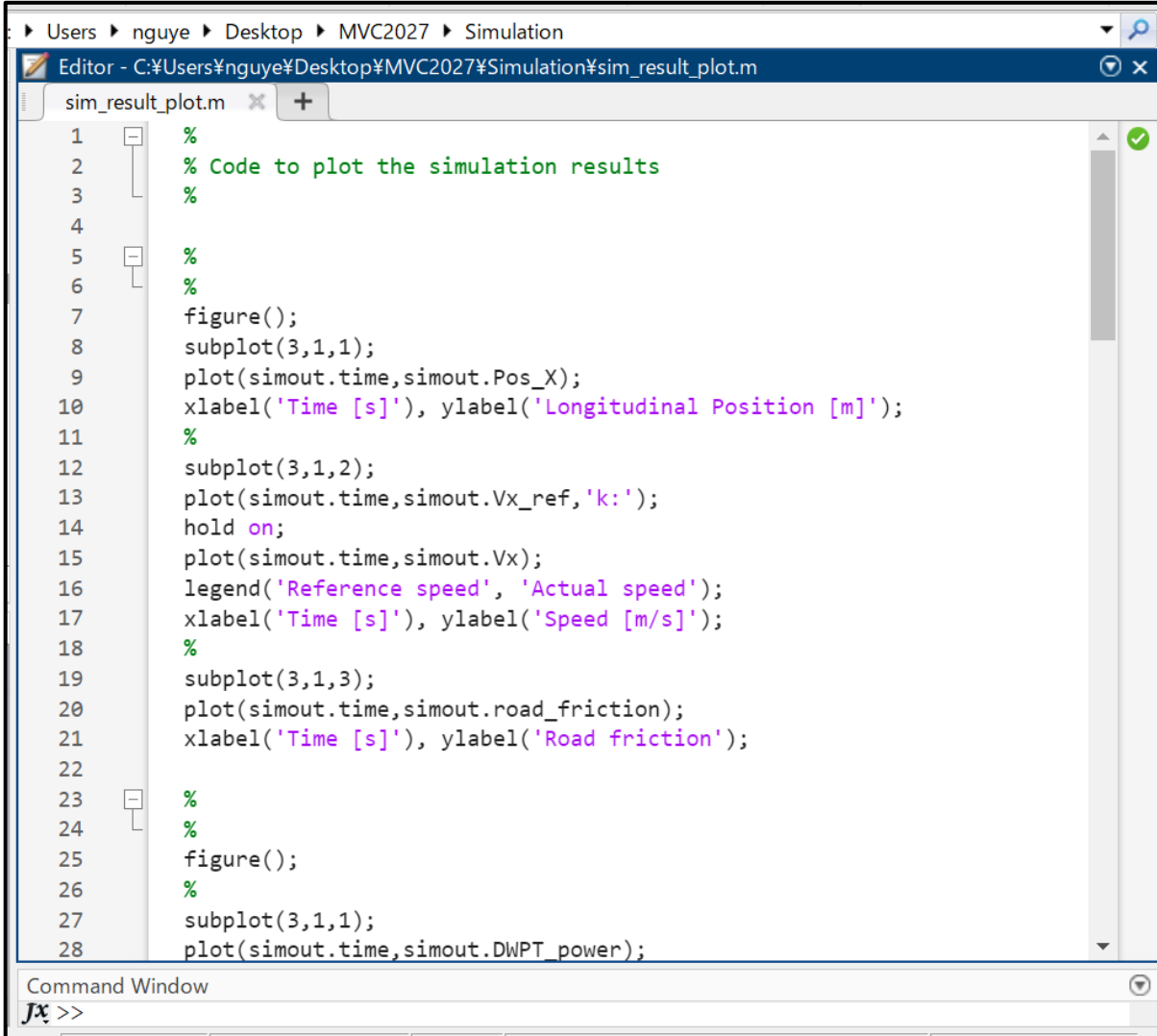
```
C:\Users\nguye\Desktop\MVC2027\Simulation
Editor - C:\Users\nguye\Desktop\MVC2027\Simulation\LambdaMethod.m
LambdaMethod.m
1 function [sys,x0,str,ts] = sfuntmpl(t,x,u,flag)
2 %SFUNTMPL General M-file S-function template
3 % 2D Tire Model Based on Lambda-Method.
4 % This S-Function is to calculate the
5 % Longitudinal (Driving) Force Fx
6 % and Lateral Force Fy
7 %
8 % based on the Magic Formula, proposed by H. B. Pacejka
9 % in "Tire and Vehicle Dynamics,"
10 % Elsevier Science, 2005.
11 %
12 % and based on the "Lambda Method", proposed by Y. Horiuchi.
13 % in "A Proposition of the Simple Tire Model for the Vehicle Stability
14 % Assist System," Proceedings of Spring Congress
15 % of Society of Automotive Engineers of Japan, No. 64-98 (1998).
16 %
17 switch flag,
18
19 %%%%%%%%%%%%%%%
20 % Initialization %
21 %%%%%%%%%%%%%%%
22 case 0,
23     [sys,x0,str,ts]=mdlInitializeSizes;
24
25 %%%%%%%%%%%%%%%
26 % Derivatives %
27 %%%%%%%%%%%%%%%
28 case 1,
```

- ❑ “**LambdaMethod.m**” is an S-Function that calculates the longitudinal and lateral forces at each wheel using Pacejka’s Magic Formula.



Example of lateral tire force characteristic

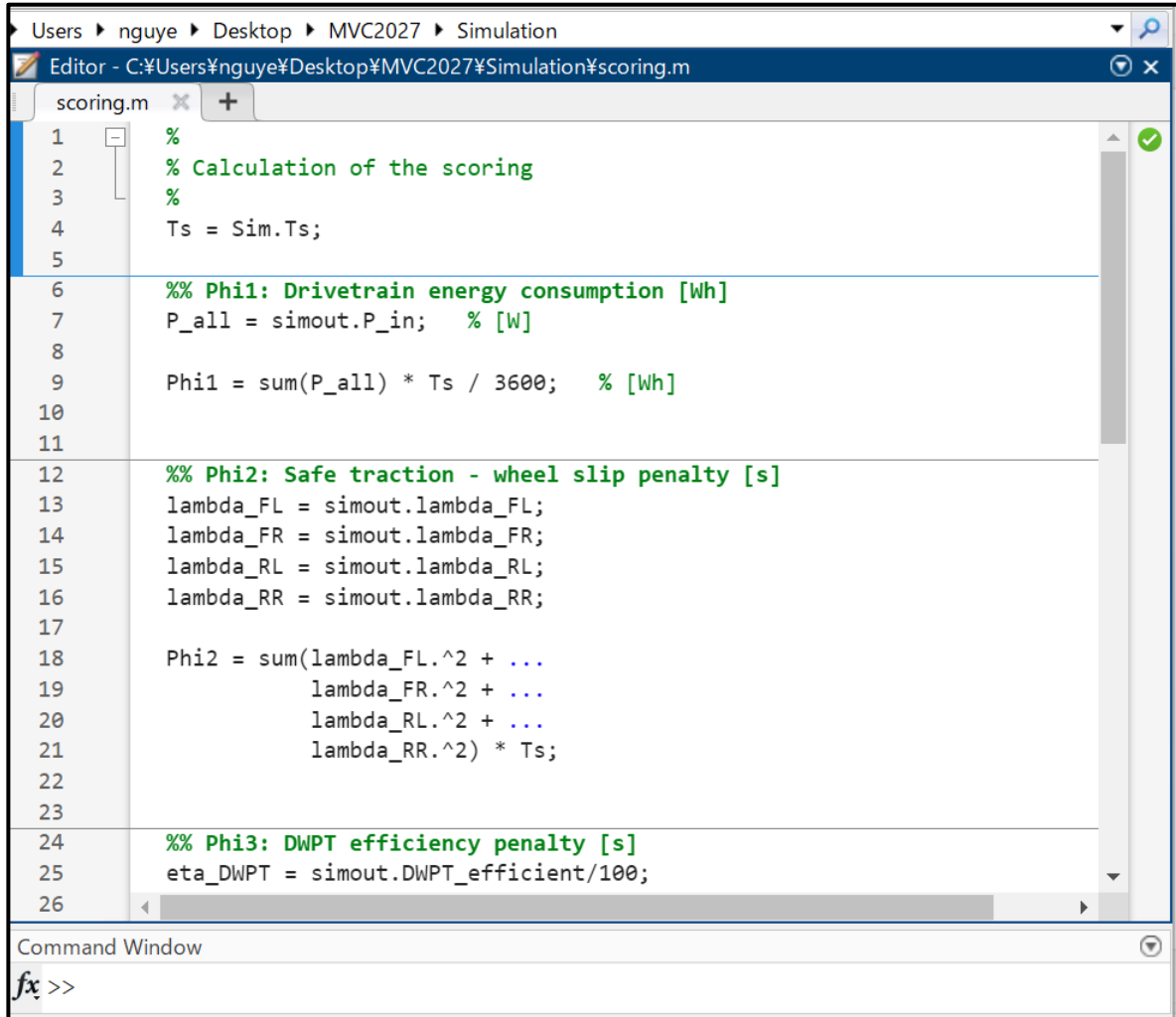
sim_result_plot.m



```
1 %  
2 % Code to plot the simulation results  
3 %  
4  
5 %  
6 %  
7 figure();  
8 subplot(3,1,1);  
9 plot(simout.time,simout.Pos_X);  
10 xlabel('Time [s]'), ylabel('Longitudinal Position [m]');  
11 %  
12 subplot(3,1,2);  
13 plot(simout.time,simout.Vx_ref,'k:');  
14 hold on;  
15 plot(simout.time,simout.Vx);  
16 legend('Reference speed', 'Actual speed');  
17 xlabel('Time [s]'), ylabel('Speed [m/s]');  
18 %  
19 subplot(3,1,3);  
20 plot(simout.time,simout.road_friction);  
21 xlabel('Time [s]'), ylabel('Road friction');  
22  
23 %  
24 %  
25 figure();  
26 %  
27 subplot(3,1,1);  
28 plot(simout.time,simout.DWPT_power);
```

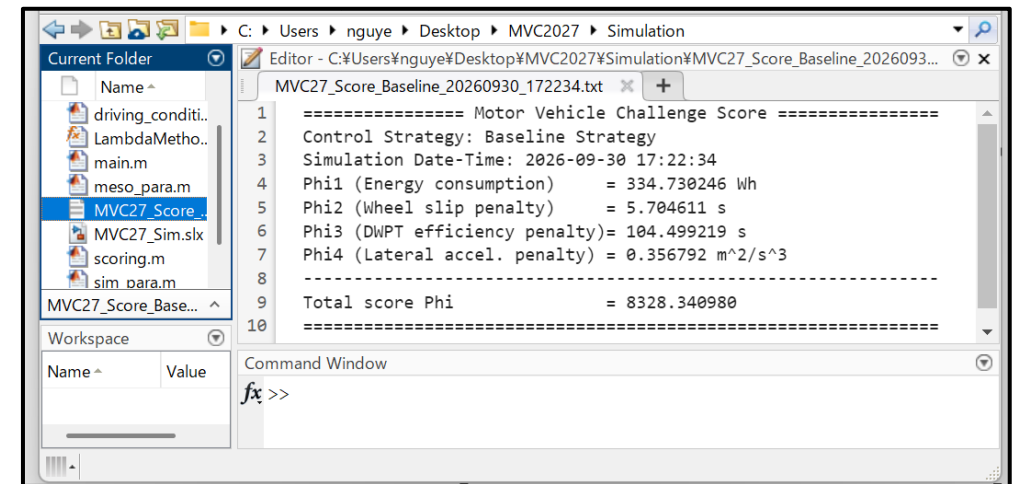
- ❑ “**sim_result_plot.m**” is to plot the main simulation results, such as
 - ✓ vehicle velocity
 - ✓ road friction coefficient
 - ✓ wheel velocities
 - ✓ lateral acceleration
 - ✓ yaw rate
 - ✓ lateral misalignment
 - ✓ efficiency of DWPT
 - ✓ energy consumed by motors
 - ✓ motor d-axis currents
 - ✓ motor torques
 - ✓ front steering angle

scoring.m



```
1 %  
2 % Calculation of the scoring  
3 %  
4 Ts = Sim.Ts;  
5  
6 %% Phi1: Drivetrain energy consumption [Wh]  
7 P_all = simout.P_in; % [W]  
8  
9 Phi1 = sum(P_all) * Ts / 3600; % [Wh]  
10  
11  
12 %% Phi2: Safe traction - wheel slip penalty [s]  
13 lambda_FL = simout.lambda_FL;  
14 lambda_FR = simout.lambda_FR;  
15 lambda_RL = simout.lambda_RL;  
16 lambda_RR = simout.lambda_RR;  
17  
18 Phi2 = sum(lambda_FL.^2 + ...  
19             lambda_FR.^2 + ...  
20             lambda_RL.^2 + ...  
21             lambda_RR.^2) * Ts;  
22  
23  
24 %% Phi3: DWPT efficiency penalty [s]  
25 eta_DWPT = simout.DWPT_efficient/100;  
26
```

- ❑ “scoring.m” is used to calculate the score and display the results in the MATLAB Command Window.
- ❑ The score is saved in a text file named `MVC27_Score_StrategyName_Date_Time.txt`
 - StrategyName is Baseline or Improved.
 - Date_Time indicates the date and time at which the simulation was executed.

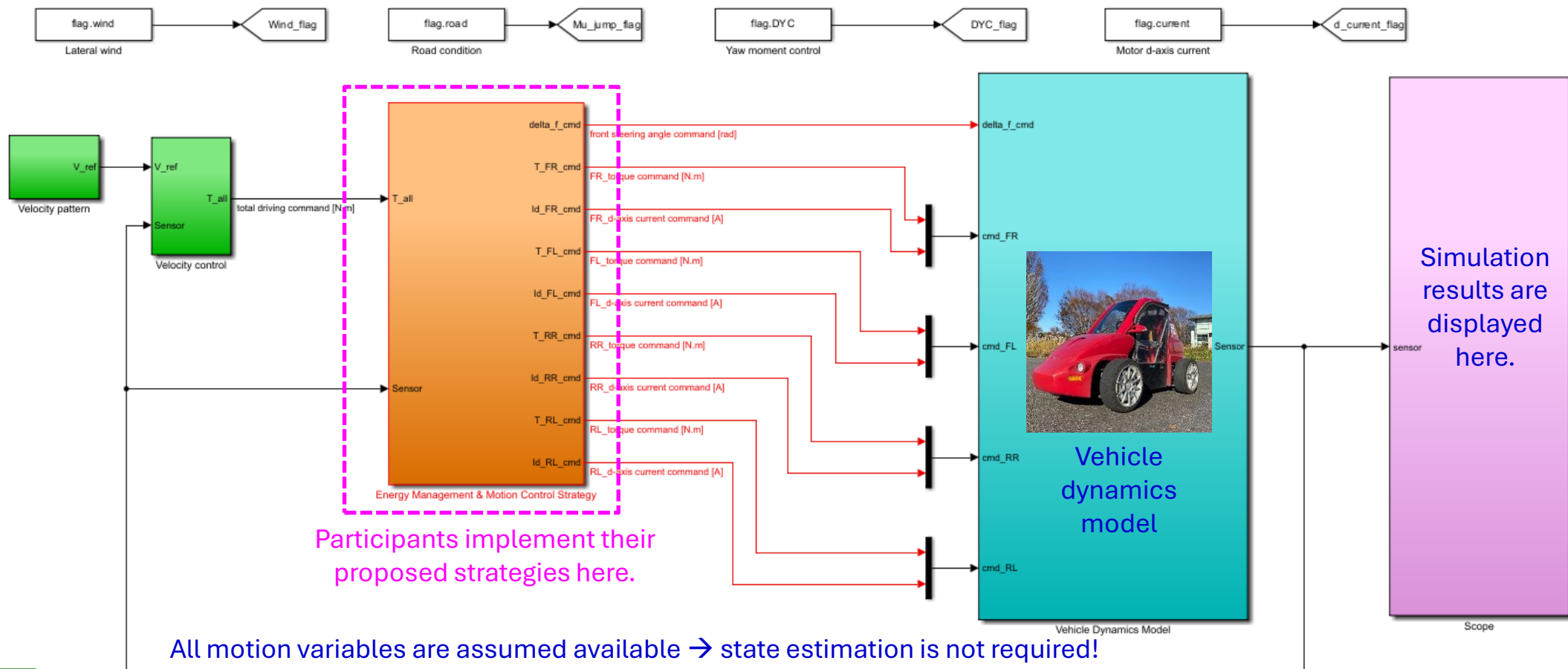


```
1 ===== Motor Vehicle Challenge Score =====  
2 Control Strategy: Baseline Strategy  
3 Simulation Date-Time: 2026-09-30 17:22:34  
4 Phi1 (Energy consumption) = 334.730246 Wh  
5 Phi2 (Wheel slip penalty) = 5.704611 s  
6 Phi3 (DWPT efficiency penalty)= 104.499219 s  
7 Phi4 (Lateral accel. penalty) = 0.356792 m^2/s^3  
8 -----  
9 Total score Phi = 8328.340980  
10 =====
```

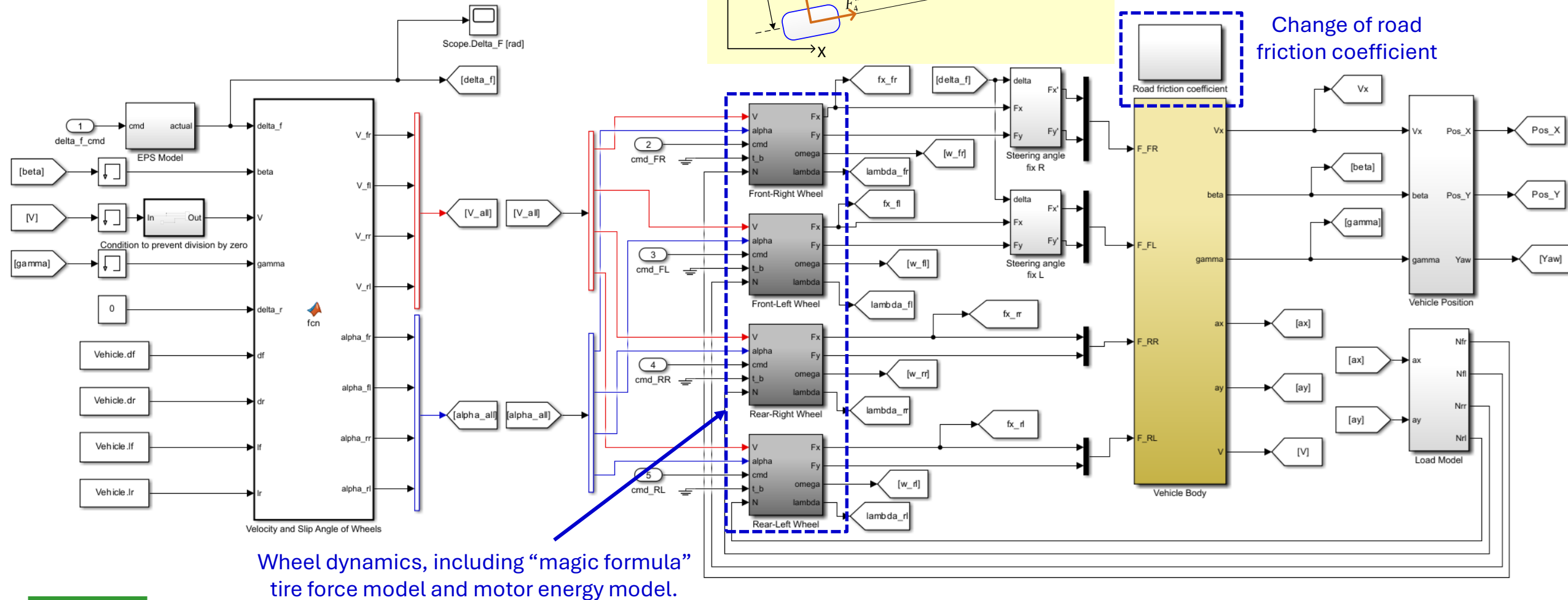
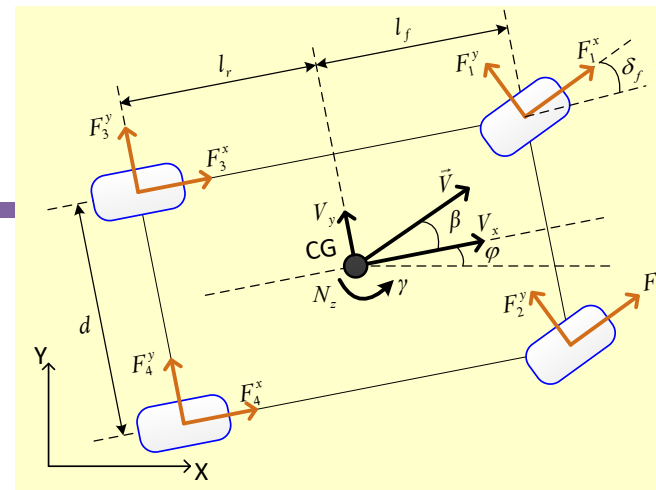
Outline of MVC27_Sim.slx

Energy Management and Motion Control of an Electric Vehicle driven by Four In-Wheel-Motors with Active Front Steering on Dynamic Wireless Power Transfer Road

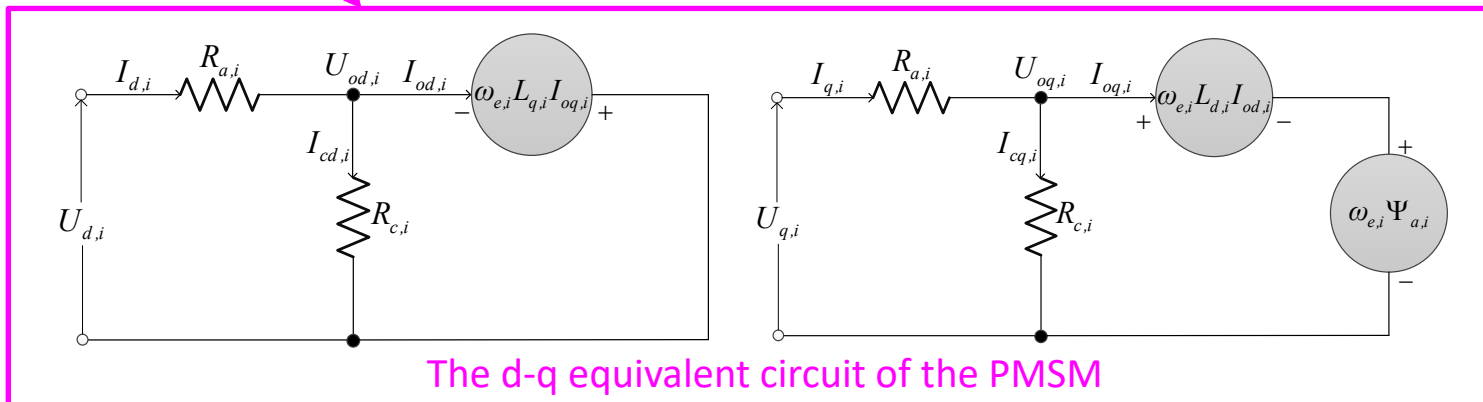
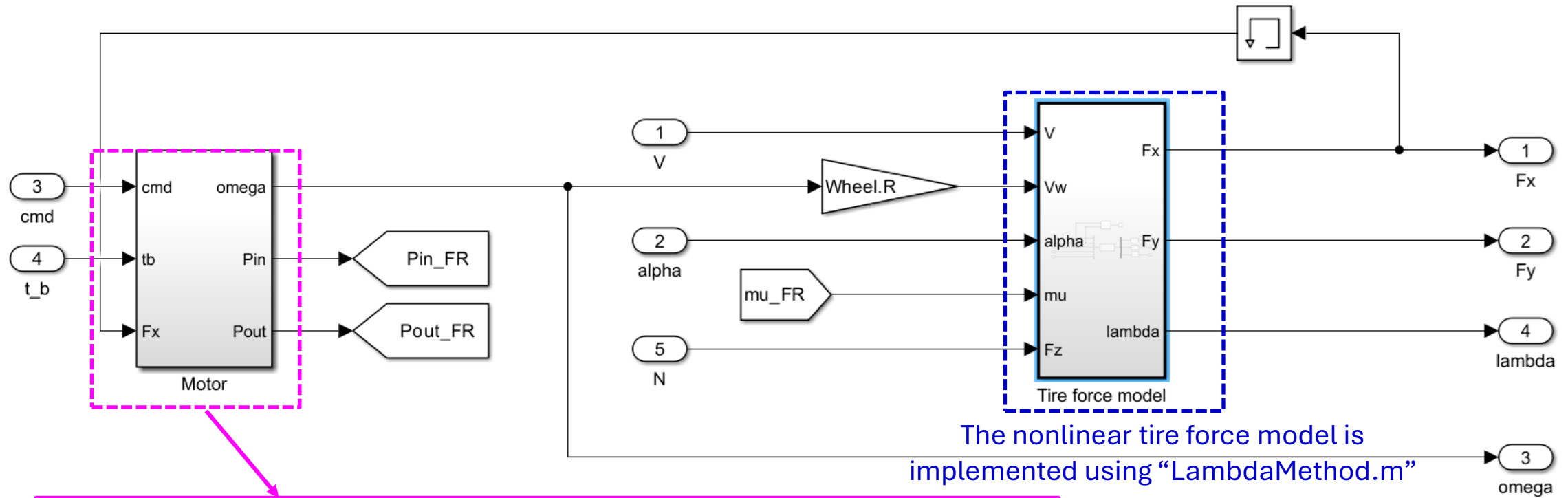
(The flags are used to set up the simulation condition)



Vehicle dynamics model



Wheel dynamics



Agenda

- ❑ Introduction
- ❑ Motivation
- ❑ Problem setting of MVC2027
- ❑ MATLAB/Simulink files
- ❑ **Demonstration of MESO strategies**
- ❑ GitHub repository
- ❑ Registration and Submission

Examples of MESO strategies

Baseline strategy (BS):

❖ Front EPS angle control:

$$\delta_f^* = K_{\delta y, p} e_y + K_{\delta \varphi, p} e_\varphi$$

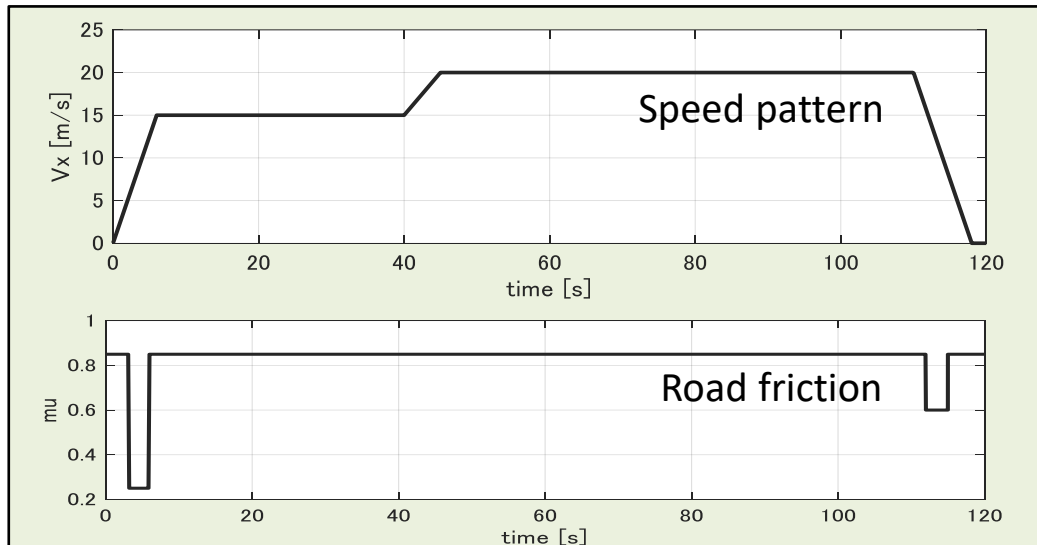
where $e_y = y^* - y$, $e_\varphi = \varphi^* - \varphi$ with the reference values $y^* = 0$ and $\varphi^* = 0$. The control gains are $K_{\delta y, p}$ and $K_{\delta \varphi, p}$.

❖ IWM torque distribution:

The torque distribution ratios are fixed as: $\{t_{1,2} = 0.27; t_{3,4} = 0.23\} \rightarrow$ no yaw moment is applied.

❖ Motor control:

The d-axis current is $I_{d,i}^* = 0$ for all IWMs.



Improved strategy (IS):

❖ Front EPS angle control:

The same as that of the BS.

❖ IWM torque distribution:

The IWM torques are distributed according to a yaw moment control law:

$$\begin{cases} T_{1,3}^* = \frac{T_{all}^*}{4} - \frac{r}{2d} N^{z*}; & T_{2,4}^* = \frac{T_{all}^*}{4} + \frac{r}{2d} N^{z*} \\ N^{z*} = K_{Ny, p} e_y + K_{Ny, i} \int_0^t e_y dt + K_{Ny, d} \frac{de_y}{dt} + K_{N\varphi, p} e_\varphi \end{cases}$$

with the control gains $K_{Ny, p}$, $K_{Ny, i}$, $K_{Ny, d}$, and $K_{N\varphi, p}$.

❖ Motor control:

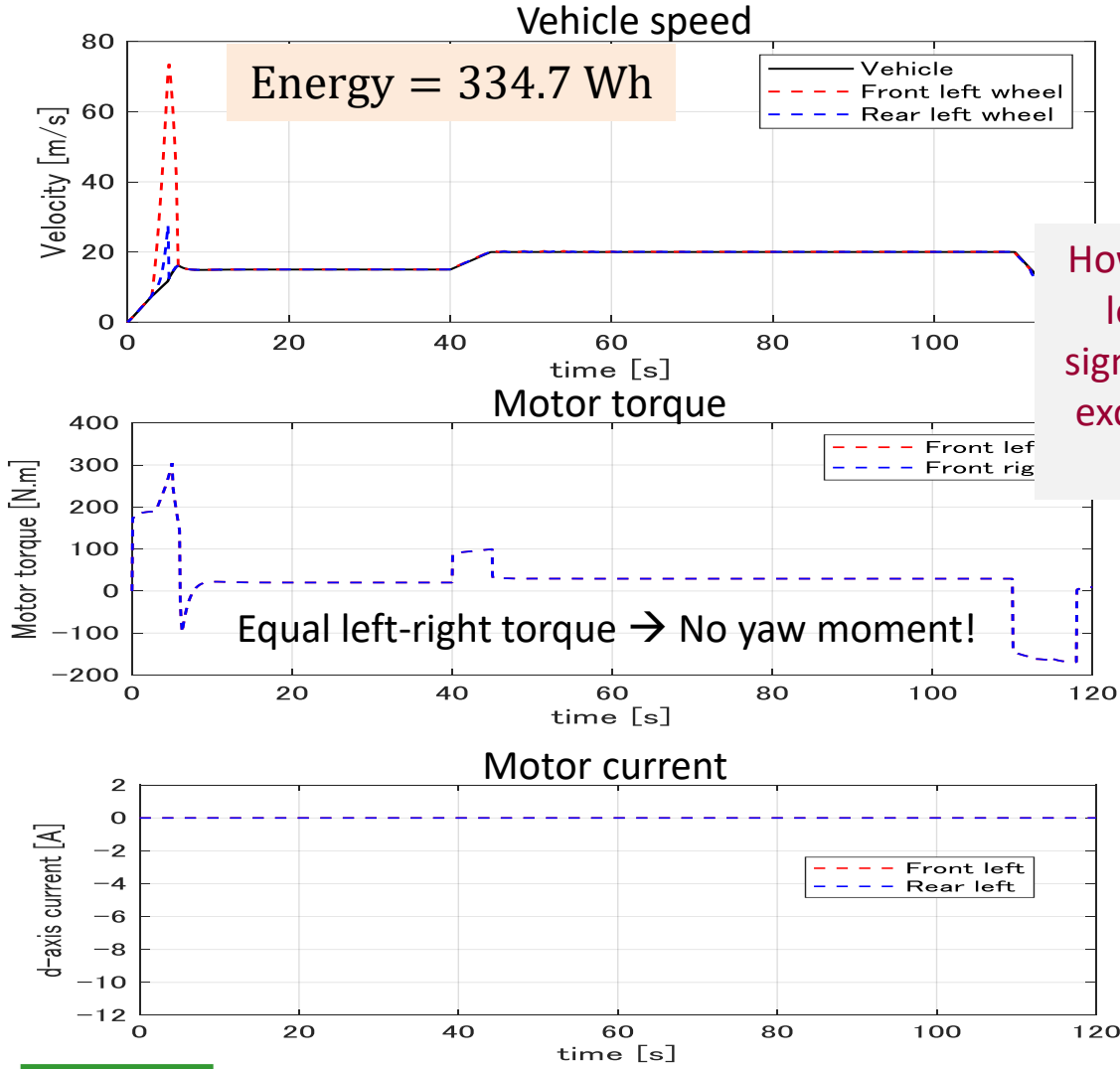
Assuming that the motor salient coefficients are close to unity, the d-axis current that minimizes motor losses is given by the following formula (Morimoto *et al*, 1994):

$$I_{od,i}^* = - \frac{\omega_{e,i}^2 L_{d,i} (R_{a,i} + R_{c,i}) \Psi_{a,i}}{R_{a,i} R_{c,i}^2 + \omega_{e,i}^2 L_{d,i}^2 (R_{a,i} + R_{c,i})}$$

S. Morimoto, Y. Tong, Y. Takeda and T. Hirasu, "Loss minimization control of permanent magnet synchronous motor drives," in *IEEE Transactions on Industrial Electronics*, vol. 41, no. 5, pp. 511-517, Oct. 1994, doi: 10.1109/41.315269.

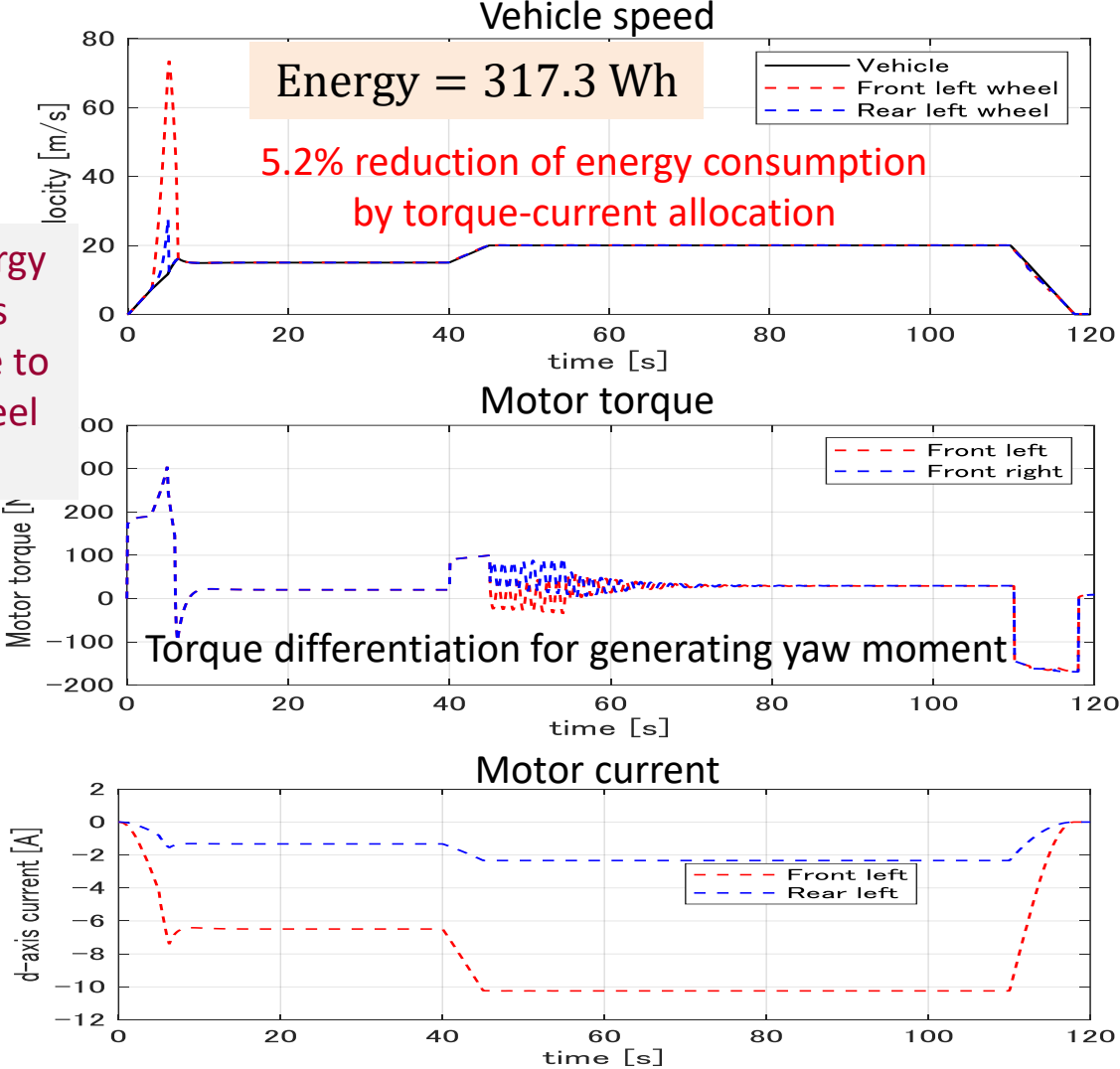
Simulation results

Baseline strategy (BS)



However, energy loss remains significant due to excessive wheel slip.

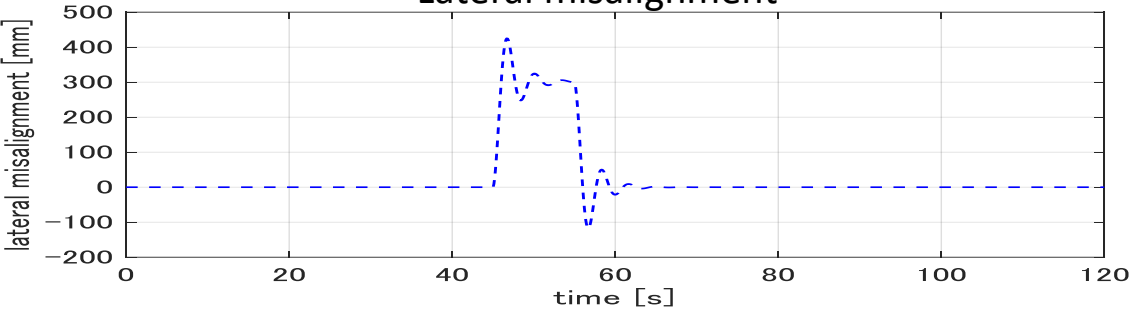
Improved strategy (IS)



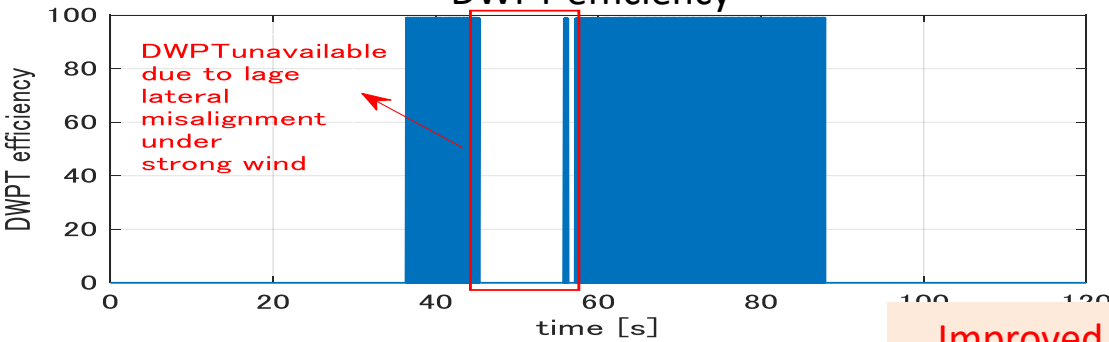
Simulation results

Baseline strategy (BS)

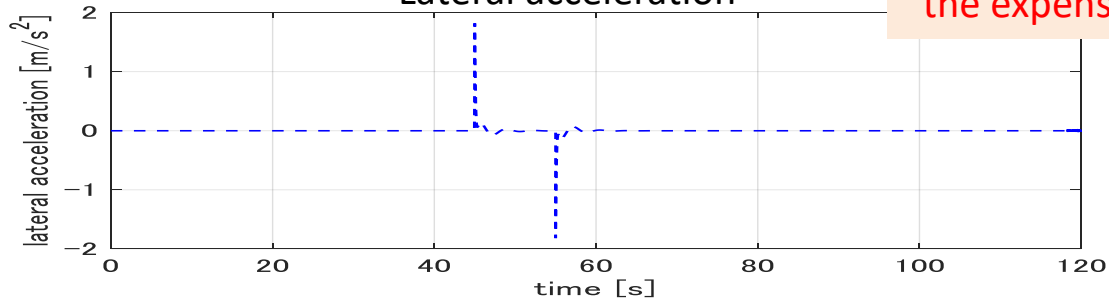
Lateral misalignment



DWPT efficiency

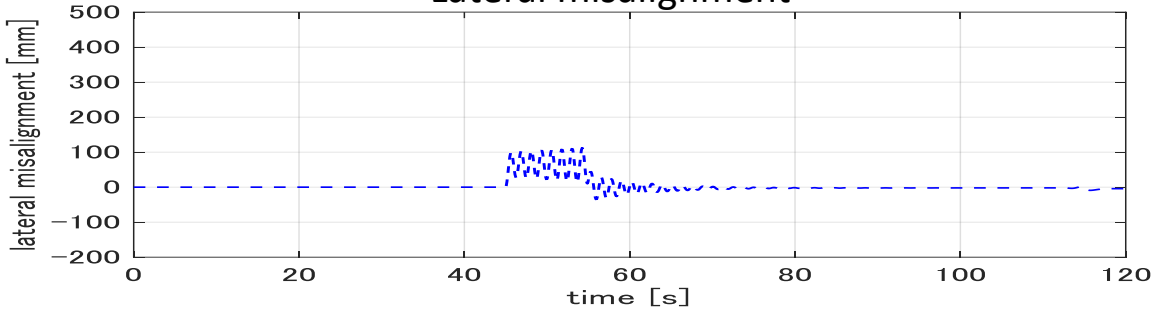


Lateral acceleration

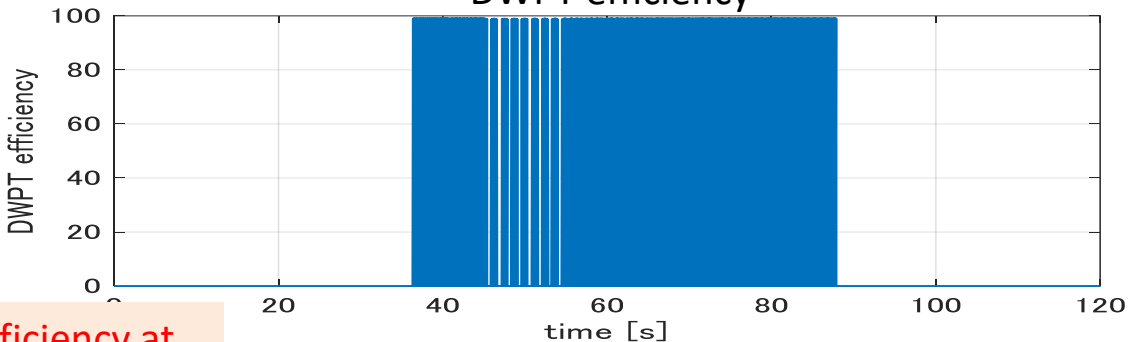


Improved strategy (IS)

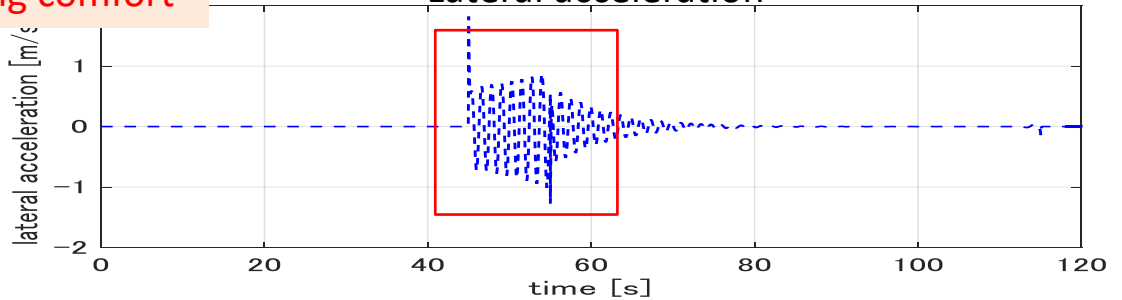
Lateral misalignment



DWPT efficiency



Lateral acceleration



Improved DWPT efficiency at the expense of driving comfort

Scoring of MESO Strategies

Baseline strategy (BS) $\phi = 994.072497$

The screenshot shows the MATLAB R2022b interface with the 'scoring.m' script open. The script contains comments for '2. Selection of the motion control & energy management' and '3. Parameter setting'. The Command Window displays the 'Motor Vehicle Challenge Score' for the Baseline Strategy, including simulation date-time, individual penalties (Phi1 to Phi4), and the total score Phi.

```
==== Motor Vehicle Challenge Score =====
Control Strategy: Baseline Strategy
Simulation Date-Time: 2026-10-04 19:41:28
Phi1 (Energy consumption) = 334.730246 Wh
Phi2 (Wheel slip penalty) = 5.704611 s
Phi3 (DWPT efficiency penalty)= 104.499219 s
Phi4 (Lateral accel. penalty) = 0.356792 m^2/s^3
-----
Total score Phi = 994.072497
=====
Score file saved as: MVC27_Score_Baseline_20261004_194128.txt
fx >>
```

Improved strategy (IS) $\phi = 968.530144$

The screenshot shows the MATLAB R2022b interface with the 'scoring.m' script open. The script contains comments for '2. Selection of the motion control & energy management' and '3. Parameter setting'. The Command Window displays the 'Motor Vehicle Challenge Score' for the Improved Strategy, including simulation date-time, individual penalties (Phi1 to Phi4), and the total score Phi.

```
==== Motor Vehicle Challenge Score =====
Control Strategy: Improved Strategy
Simulation Date-Time: 2026-10-04 20:00:04
Phi1 (Energy consumption) = 317.270407 Wh
Phi2 (Wheel slip penalty) = 5.395927 s
Phi3 (DWPT efficiency penalty)= 102.188848 s
Phi4 (Lateral accel. penalty) = 2.093757 m^2/s^3
-----
Total score Phi = 968.530144
=====
Score file saved as: MVC27_Score_Improved_20261004_200004.txt
fx >>
```

Score Analysis

Metric	Baseline Strategy (BS)	Improved Strategy (IS)	Change (IS – BS)	Change
Φ_1 : Energy consumption	334.730246 Wh	317.270407 Wh	–17.459839 Wh	–5.22%
Φ_2 : Wheel slip penalty	5.704611 s	5.395927 s	–0.308684 s	–5.41%
Φ_3 : DWPT efficiency penalty	104.499219 s	102.188848 s	–2.310371 s	–2.21%
Φ_4 : Lateral accel. penalty	0.356792 m ² /s ³	2.093757 m ² /s ³	+1.736965 m ² /s ³	+486.83%
Total score Φ	994.072497	968.530144	–25.542353	–2.57%

% 4. Weighting of scoring function
sigma = [12 200 30 100];

The improved strategy reduces the total score by **2.57%**, mainly through reductions in energy consumption, wheel slip, and DWPT efficiency penalty, while sacrificing lateral-acceleration performance.

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- ❑ Registration and Submission

GitHub Repository

It contains:

- Introduction to MVC2027
- Simulation files
- Deadlines
- Competition rules
- Registration link
- Submission link
- Any other relevant information/materials
 - MVC2027 paper
 - paper presentation slide
 - workshop slide

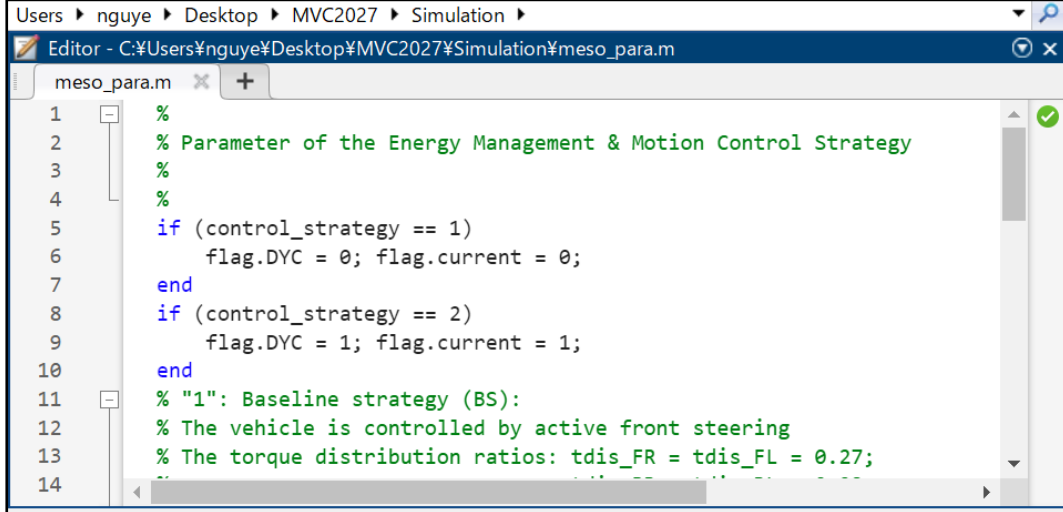
https://github.com/VTsociety/MVC_2027/



- Registration deadline: 31 January 2027
- Submission: 1 March 2027
- Results: 31 March 2027

Competition Rules

1. The MATLAB version used for the submitted files must be **R2022b**.
2. Participants are allowed to modify only the “**meso_para.m**” script and the “**Energy Management & Motion Control Strategy**” block provided in the simulation file.
3. All parameters in “**sim_para.m**”, as well as all measurements available from the output of the “Vehicle Dynamics Model” block, may be used to develop the proposed strategy.
4. All other scripts and simulation blocks **must not** be modified. Any modifications made to scripts other than “**meso_para.m**” or to subsystems other than “**Energy Management & Motion Control Strategy**” will be discarded during the evaluation.
5. The final score will be calculated as the average of the scores obtained using the example and secret driving cycles.
6. A proposed strategy will be disqualified if the vehicle *fails to complete* either driving cycle (e.g., the vehicle stops before reaching the end of the driving cycle), or if the *lateral position of the vehicle exceeds 5 m* at any time during the simulation.



```
1 %  
2 % Parameter of the Energy Management & Motion Control Strategy  
3 %  
4 %  
5 if (control_strategy == 1)  
6     flag.DYC = 0; flag.current = 0;  
7 end  
8 if (control_strategy == 2)  
9     flag.DYC = 1; flag.current = 1;  
10 end  
11 % "1": Baseline strategy (BS):  
12 % The vehicle is controlled by active front steering  
13 % The torque distribution ratios: tdis_FR = tdis_FL = 0.27;  
14 %
```

Remark 1:

- ❖ Modify “**meso_para.m**” so that the proposed strategy is selected and executed in the simulation when “**control_strategy == 2**”.

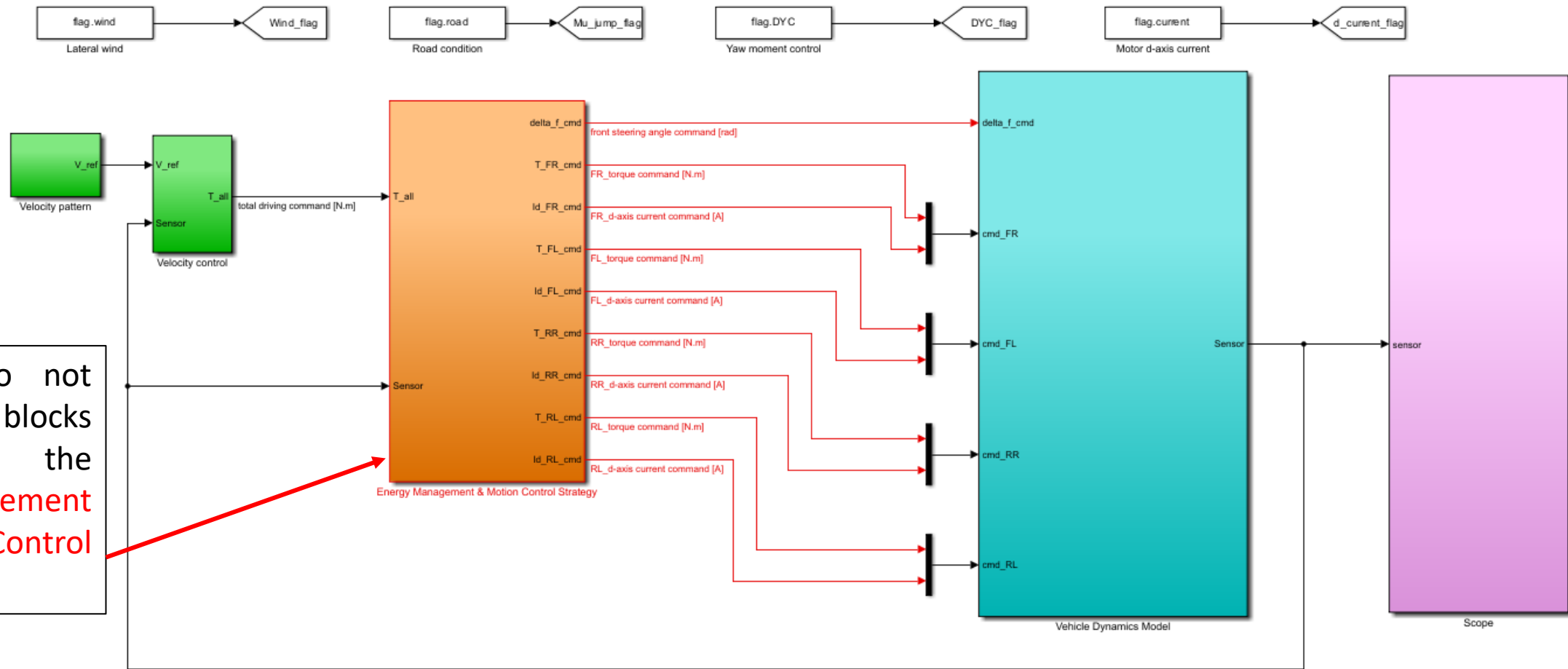
Competition Rules

```
27 %
28 %
29 %
30 % 1. Active front steering
31 Steer.Kp_pos = 0.02; % P gain of the lateral position control via front steering
32 Steer.Kp_yaw = 0.2; % P gain of the yaw angle control control via front steering
33 %
34 % 2. Yaw moment control (Improved strategy)
35 DYC.Kp_pos = 4000; % P gain of lateral position control via DYC
36 DYC.Ki_pos = 20; % I gain of lateral position control via DYC
37 DYC.Kd_pos = 1000; % D gain of lateral position control via DYC
38 DYC.Taud_pos = 0.005; % Time constant of Derivative terms
39 DYC.Kp_yaw = 5000; % P gain of yaw angle control via DYC
40 %
41 % 3. Torque distribution (Baseline strategy)
42 tdis_FR = 0.27; % torque distribution gain (front right)
43 tdis_FL = 0.27; % torque distribution gain (front left)
44 tdis_RR = 0.23; % torque distribution gain (rear right)
45 tdis_RL = 1 - (tdis_FR + tdis_FL + tdis_RR); % torque distribution gain (rear left)
46 %
```

Remark 2:

- ❖ Participants are free to modify “**meso_para.m**” as needed to implement their proposed strategies.
- ❖ The current parameters are provided only as examples. Participants are free to modify the control gains or develop alternative methods for **steering angle control, torque distribution, and d-axis current control**.
- ❖ The proposed strategy *must not* directly access information and parameters defined in LambdaMethod.m and driving_condition.m. The strategy may use only the parameters explicitly permitted by the rules and the available sensor measurements to estimate/predict the unknown parameters.

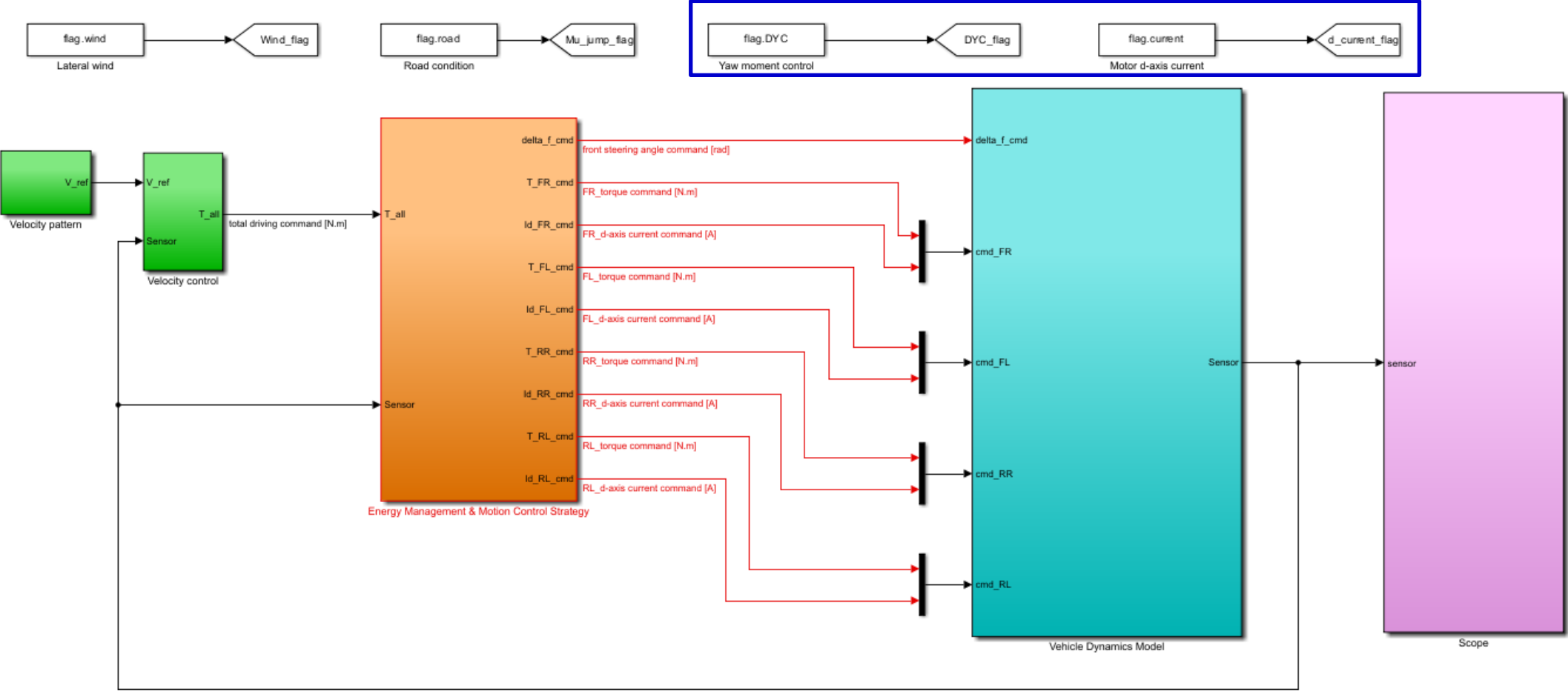
Competition Rules



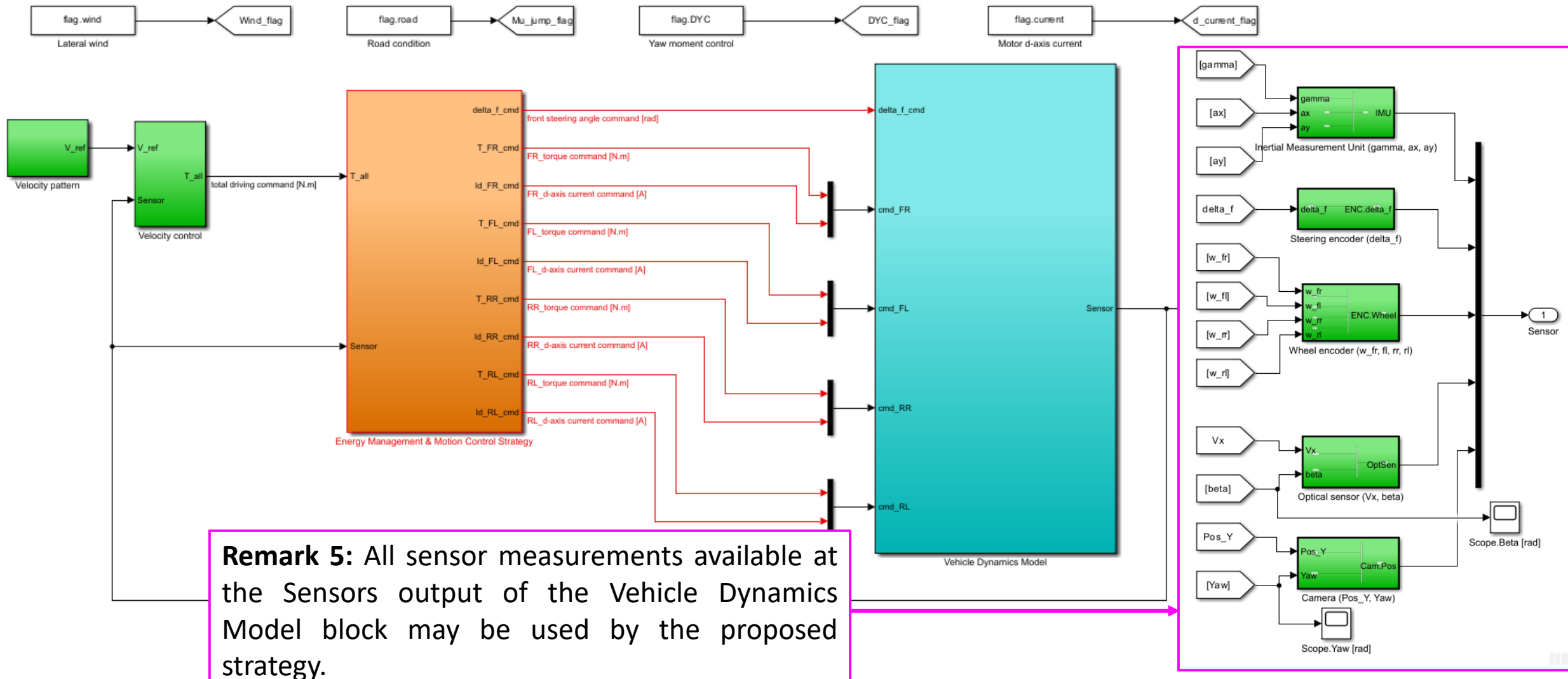
Remark 3: Do not modify any blocks other than the “Energy Management & Motion Control Strategy” block.

Competition Rules

Remark 4: The “DYC_flag” and “d_current_flag” may be used and modified as needed. Additional flags may also be defined in accordance with the proposed control strategy.



Competition Rules



Competition Rules

Remark 6: For each driving cycle, the score will be calculated as the sum of the scores obtained under the following driving conditions:

- ❖ **Condition 1** {flag.wind = 0; flag.road = 0}: high friction surface *without* lateral wind disturbance
- ❖ **Condition 2** {flag.wind = 1; flag.road = 0}: high friction surface *with* lateral wind disturbance
- ❖ **Condition 3** {flag.wind = 0; flag.road = 1}: friction variation surface *without* lateral wind disturbance
- ❖ **Condition 4** {flag.wind = 1; flag.road = 1}: friction variation surface *with* lateral wind disturbance

$$\phi_{\text{example}} = \sum_{j=1}^4 \phi_{\text{example}}(\text{jth condition})$$

$$\phi_{\text{secret}} = \sum_{j=1}^4 \phi_{\text{secret}}(\text{jth condition})$$



$$\phi_{\text{final}} = \frac{\phi_{\text{example}} + \phi_{\text{secret}}}{2}$$

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Registration

IEEE VTS Motor Vehicle Challenge 2027 (MVC2027) - Resigtration Form

nguyen.binhminh@edu.k.u-tokyo.ac.jp アカウントを切り替える

* 必須の質問です

メール *

☒ 返信に表示するメールアドレスとして nguyen.binhminh@edu.k.u-tokyo.ac.jp を記録する

Team Name *

回答を入力

Name of Member 1 (contact person) *

回答を入力

Surname of Member 1 (contact person) *

Check this box
to receive the
confirmation
email.

Please enter the
team name,
contact email
address, and
names of all
team members,
and answer all
the questions.

Participants
must submit the
registration
form before
submitting the
proposed
strategies.

https://docs.google.com/forms/d/e/1FAIpQLScn9QM5dTbPJO58kUCpS6MpwM0aLW2bU7g9Atnztza7pp8_2uQ/viewform

What has to be uploaded for evaluation?

In order to evaluate each proposal, the teams must submit in the form the following files:

- ❖ “meso_para.m” renamed as “nameTeam_numProp_meso_para.m” where “numProp” must identify the proposal number of the team. If only one proposal is presented, please use “num1” (e.g., “teamABC_prop1_meso_para.m”). Please avoid spaces in the file name.
- ❖ “MVC27_Sim.slx” renamed as “nameTeam_numProp_MVC27_Sim.slx” where “numProp” must identify the proposal number of the team. If only one proposal is presented, please use “num1” (e.g., “teamABC_prop1_MVC27_sim.slx”). Please avoid spaces in the file name.
- ❖ Complete simulation package of the proposed solution (*backup*): Create a ZIP file including the following files: main.m, sim_para.m, nameTeam_numProp_meso_para.m, driving_condition.m, LambdaMethod.m, sim_result_plot.m, scoring.m, nameTeam_numProp_MVC27_Sim.slx.
- ❖ The score text file may also be included. Participants can also include a file that briefly explains their idea, and any instruction for the organizers to reproduce and evaluate the proposed strategy.
- ❖ Rename the ZIP file as “nameTeam_numProp_MVC27.zip”. Please note the ZIP file is only a backup.
Evaluation will mainly be based on the individual submitted files.

Submission

Check this box to receive the confirmation email.

Please enter the team name, name of the contact person, and contact email address.

Upload files from your computer

- nameTeam_numProp_meso_para.m
- nameTeam_numProp_MVC27_Sim.slx
- nameTeam_numProp_MVC27.zip

Click this button to submit files

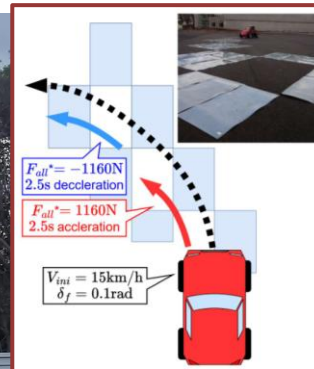
送信

フォームをクリア

<https://docs.google.com/forms/d/e/1FAIpQLSdaAZOpIFS4j9ZSh1LC5OPYZo41tUQKE3v68uK8hWsC88eRkA/viewform>

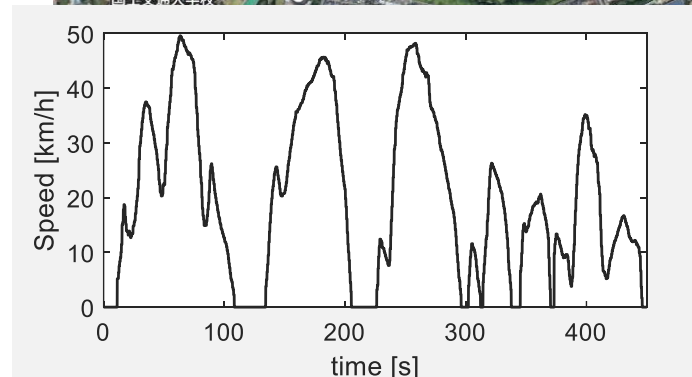
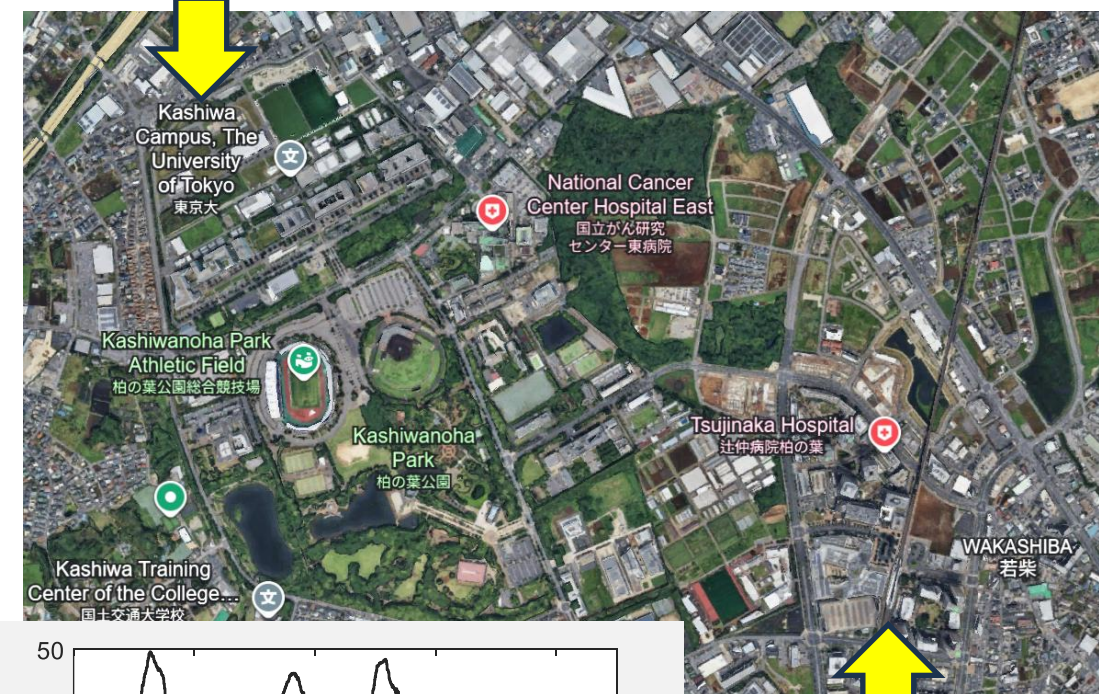
Secret driving cycle

- ❖ The secret driving cycle is established based on a **real-world driving pattern** recorded for a vehicle traveling from the University of Tokyo to Kashiwanoha-campus Station.
- ❖ The road friction varies randomly and non-uniformly between the left and right wheels.
- ❖ The vehicle may encounter low-friction surfaces and lateral wind disturbances simultaneously, including during acceleration and deceleration.



Example of road surface with random friction

The University of Tokyo



Kashiwanoha Campus Station

Questions and Support

- ❑ Please post technical questions on GitHub.
- ❑ Please do not hesitate to contact us by email if you have any questions.

Binh-Minh Nguyen: nguyen.binhminh@edu.k.u-tokyo.ac.jp

Facundo Aguilera: faguilera@ing.unrc.edu.ar

- ❑ We might organize online meeting to discuss with you and provide technical support.

ENJOY THE CHALLENGE AND MAY THE BEST WIN

https://github.com/VTSociety/MVC_2027/



- Registration deadline: 31st January 2027
- Submission: 1st March 2027
- Results: 31 March 2027

**IEEE****VTS**
Connecting the Mobile World

IEEE Vehicular Technology Society

IEEE VTS Motor Vehicle Challenge 2027

**Management of Energy Efficiency and Safe Operation
for Multi-Actuated Ground Vehicles on DWPT Road**

The Motor Vehicle Challenge (MVC), supported by IEEE Vehicular Technology Society, is an annual activity to find an appropriate energy management strategy to improve electric vehicles' performance. In this eleventh edition of the event, the challenge focuses on the energy-efficient and safe operation of an in-wheel-motor vehicle with dynamic wireless power transfer operating on a *road subject to various disturbances*. The competition is open to everyone, students, academics, and industry.

DWPT CHARGING COILS

CHALLENGE

This challenge considers a four in-wheel-motor vehicle (4IWM-EV) operating autonomously on roads with a dynamic wireless power transfer (DWPT) infrastructure, based on prototypes developed at the University of Tokyo. Under realistic conditions, the vehicle is subjected to abrupt variations in road friction due to weather changes, as well as strong lateral wind disturbance. Participants are therefore invited to develop an integrated strategy that coordinates motor torque distribution, d-axis current control, and front steering angle.

The MVC2027 intends to serve as a platform for discussing and advancing the concept of MESO (Management of Energy Efficiency and Safe Operation) for multi-actuated vehicles.

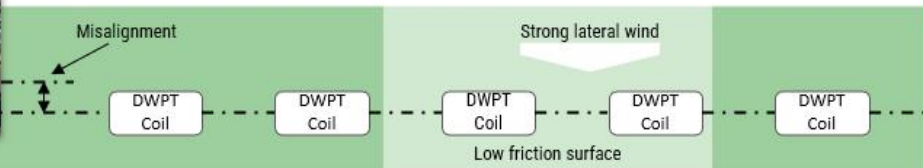
SCORING

Participants of the Challenge will be provided with a model of the 4IWM-EV and DWPT platform on MATLAB/Simulink, together with sample control strategies and speed profiles. Each proposed strategy will be evaluated using a scoring function that assesses the achievement of multiple objectives, including:

- improving overall drivetrain efficiency;
- ensuring safe traction by regulating wheel slip ratios;
- enhancing infrastructure utilization by maximizing the energy received from DWPT;
- improving driving comfort by minimizing deviations in lateral acceleration.



4IWM-EV



Misalignment

Strong lateral wind

Low friction surface

DWPT Coil